

Legal Process and Requirements for Cloud Forensic Investigations

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ABSTRACT

For the emerging field of cloud forensics, the development of validated and repeatable scientific processes for conducting cloud forensic investigations should include requirements that establish evidence collected is legally admissible. There is currently an uncertainty in the legal requirements for cloud forensics. Forensic investigations in the cloud introduce unique issues that must be addressed, and the legal environment of the cloud must be considered. The authors will detail the process in criminal cloud forensic investigations for commanding production from cloud providers including constitutional and statutory limitations; and the civil and criminal admissibility processes. Decisions in court cases rely on the authenticity and reliability of the evidence presented. Ensuring cases involving cloud forensics follow the proper legal process and requirements will be beneficial for validating evidence when presented in court. Further, understanding of legal requirements will aid in the research and development of cloud forensics tools to aid investigations.

1.0 INTRODUCTION

Cloud forensics introduces unique legal issues beyond those encountered during traditional digital forensics cases and presents a challenge to the legal system which is not well equipped to handle such cases. This chapter will examine issues regarding commanding and producing in court digital evidence resident in the cloud. The "commanding production" section will focus on the criminal law. The "producing in court" or admissibility section will apply to both civil and criminal practice.¹

¹ The decision to limit the discussion regarding methods of commanding production to criminal investigations flows primarily from two differences between criminal and civil practice: First, there are constitutional limits on the acquisition of evidence in criminal investigations that are not applicable to civil investigations.

Second, in general, the investigative stage of a criminal case is largely complete by the time the case is filed. "Discovery" obligations in criminal cases largely refer to the constitutional and statute/rule based obligations of the parties to disclose their evidence to the other side.

To date, there is limited guidance available from case law that can govern decisions involving admissibility of cloud-based evidence. Our analysis is founded on an extensive review of the constitutional and statutory limitations that apply to cloud forensic investigations, as well as a walkthrough of admissibility standards for digital evidence including issues unique to cloud-based evidence.

Cloud computing is in its infancy. This chapter identifies the ways in which digital evidence in the cloud differs in substance from digital evidence gathered from computer hard drives and networks under the control of parties engaged in legal actions. The material presented begins to identify the issues surrounding cloud forensics uncertainty, comparing these issues to those raised in more traditional digital forensics cases.

The authors recognize that addressing barriers to conducting effective cloud forensic investigations will require a concerted effort by stakeholders involved in the process, including the cloud provider.

Development of new tools and procedures for cloud forensics may not currently address the complex legal requirements that must be met in order for cloud-gathered evidence to be admissible in court. Incorporating the need to collect admissible evidence in system design can improve the ability of system operators to identify, collect, store and retrieve valid evidence. It is particularly important for potential cloud customers to analyze this before moving to the cloud since cloud customers lose the ability to control this process once information is moved to the cloud.² Understanding the legal requirements for admissible cloud evidence allows for incorporation of those concepts into information systems, creating a forensically ready design that will improve the efficiency of valid evidence collection.

1.1 Cloud Forensics Definitions

The emerging field of cloud forensics combines the disciplines of digital forensics and cloud computing.³ Cloud computing is defined by the U.S. National Institute of Standards and Technology (NIST) as “a model for enabling convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction”.⁴

In contrast, the investigative stage frequently only starts when a civil case has been filed. Discovery is an affirmative and aggressive process, continuing from when a case is filed to trial or any earlier discovery cutoff date. We felt that covering the acquisition of cloud data in the civil context would expand the discussion so as to make it unmanageable as a chapter in book.

² Convery, N. (2010) Cloud Computing Toolkit: Guidance for outsourcing information storage to the cloud. Department of Information Studies, Aberystwyth University, Wales, 20. Retrieved July 22, 2012, from http://www.archives.org.uk/images/documents/Cloud_Computing_Toolkit-2.pdf

³ Ruan, K., Carthy, J., Kechadi, M. T., & Crosbie, M. (2011b). CLOUD FORENSICS. IFIP Int. Conf. Digital Forensics, IFIP Advances in Information and Communication Technology, 35–46. doi:10.1007/978-3-642-24212-03

⁴ Mell, P., & Grance, T. (2011). *Definition of Cloud Computing: NIST Special Publication 800-145*. Gaithersburg, MD: Computer Security Division, Information Technology Laboratory National Institute of Standards and Technology, 2. Retrieved from <http://csrc.nist.gov/publications/nistpubs/800-145/SP800-145.pdf>

Digital forensics is the study of evidence from attacks on computer systems in order to learn what has occurred, how to prevent it from recurring, and the extent of the damage. This field initially was divided into digital disk forensics—retrieving admissible evidence from a computer disk—and network forensics—retrieving evidence throughout a network system wherever it may reside or flow. Based on network access and architecture, cloud forensics is a subset of network forensics.⁵

With traditional disk forensics, the model by which investigations are conducted rely on the acquisition of physical disks, and require a clear chain of custody be maintained on the physical items.⁶ By contrast, cloud forensics requires a different approach due to characteristics of the cloud environment where physical assets are not under the control of the user and may not be identified and located easily due to the dynamic nature of cloud provisioning.

1.2 Legal Background

Cloud forensics, like traditional digital forensics, requires that technical and legal practitioners have a strong integrated understanding of computer science and the law. U.S. courts operate through a combination of procedural rules and case law precedent as guidance when ruling on evidentiary issues.⁷ Other countries where cloud data may reside may have a different basis for legal decision-making, complicating the collection of cloud-based digital evidence

Case law precedent is based on the concept of *stare decisis*, which establishes that the court rules by using precedence to guide decisions on cases that are similar in nature.⁸ In other words, courts will make decisions on the basis of rulings that have already been decided and what will be ruled on has the potential to shape how the court rules on cases of similar nature in the future. Since the field of cloud forensics has not established best practices to the level of similar fields such as digital forensics,⁹ early rulings could be based on faulty science, complicating the development of adequate case law and setting faulty first precedent.

Our research emphasizes the procedural rules and decisions that guide the admissibility of data as valid evidence in the U.S. legal system. In that system admissibility decisions are the purview of the judge who is permitted significant discretion. While the decision lies with the judge, the burden is on opposing counsel to objecting to admissibility.¹⁰ It is our expectation that raising and examining the issues of admissibility of cloud forensics evidence in advance of significant case law being established, will avoid some of the pitfalls of rulings based on poor science.

1.3 Literature Review

With the field of cloud forensics in its infancy, little in the relevant literature details the legal aspects of admissibility of cloud forensic data; however, publications that have discussed cloud

⁵ Ruan, K., Carthy, J., Kechadi, M. T., & Crosbie, M. (2011b).

⁶ Pollitt, M., Caloyannides, M., Novotny, J., & Sheno, S. (2004). Digital forensics: operational, legal and research issues. *Data and Applications Security XVII*, 393–403.

⁷ Kuntze, N., Rudolph, C., Alva, A., Endicott-Popovsky, B., Christiansen, J. R., & Kemmerich, T. (2012). On the Creation of Reliable Digital Evidence. In S. Sheno, (Ed.), *Advances in Digital Forensics VIII*. Heidelberg: Springer.

⁸ Re, E. D. (1975). *Stare Decisis. Presented at a Seminar for Federal Appellate Judges sponsored by the Federal Judicial Center*, An exploration of the doctrine of precedent in the judicial process (pp. 1-21). Retrieved from [http://www.fjc.gov/public/pdf.nsf/lookup/staredec.pdf/\\$file/staredec.pdf](http://www.fjc.gov/public/pdf.nsf/lookup/staredec.pdf/$file/staredec.pdf)

⁹ Christiansen, J. (2010). Discovery and admission of electronic information as evidence. In J. Sullivan (Ed.), *E-Health Business and Transactional Law* (pp. 427–452). Arlington, Virginia: BNA Books.

¹⁰ *Lorraine v. Markel American Insurance Company*, 241 F.R.D. 534 (D.Md. 2007).

forensics have mentioned, in passing, that legal requirements are important to consider. Examples include: Biggs & Vidalis,¹¹ Birk & Wegener,¹² Broadhurst,¹³ ; Ruan, Carthy, Kechadi, & Crosbie,¹⁴ Spyridopoulos & Katos,¹⁵ Taylor, Haggerty, Gresty, & Hegarty (2010),¹⁶ Taylor, Haggerty, Gresty, & Lamb (2011),¹⁷ Wolthusen,¹⁸ Yan.¹⁹

2.0 COMMANDING PRODUCTION

As noted at the outset, our focus in this section will be on the criminal investigation process.

2.1 Introduction

This section discusses the laws and rules governing the issuance of legally binding directives and the circumstances under which they must be used. Within each topic these are discussed, first, in general, and then second, as these laws and rules relate to storage in the cloud. In doing so the authors will attempt to identify issues unique to, or causing special problems for, the forensic examination of cloud stored data.

For the purposes of our discussion, a "legally binding directive" is a court order that mandates compliance by the recipient and meets all necessary constitutional and statutory requirement. Two observations:

First, where the constitution or relevant statute mandates a particular form of order (such as a warrant supported by probable cause, for example), failure to utilize the proper order can lead to adverse results, such as suppression of any evidence obtained and/or civil or criminal liability for providing/obtaining such information.

Second, the absence of constitutional or statutory protection does not mean the cloud provider *must* provide a third party access to cloud data. Even where the constitution or statutes do not prevent the cloud service provider from allowing third party access, because of Service Level Agreements, Privacy Policies and other factors, it is highly unlikely that an investigator will be able to access records stored in the cloud without producing a legally binding directive to the cloud provider.

¹¹ Biggs, S., & Vidalis, S. (2009). Cloud computing: The impact on digital forensic investigations. *Internet Technology and Secured Transactions*, 2009. ICITST 2009. *International Conference for*, 1–6.

¹² Birk, D., & Wegener, C. (2011). Technical issues of forensic investigations in cloud computing environments. *Systematic Approaches to Digital Forensic Engineering (SADFE)*, 2011 *IEEE Sixth International Workshop on*, 1–10. doi:10.1109/SADFE.2011.17

¹³ Broadhurst, R. (2006). Developments in the global law enforcement of cyber-crime. *Policing: An International Journal of Police Strategies & Management*, 29(3), 408–433.

Cable Privacy Act, 47 U.S.C. § 551 (2000) and the Video Privacy Protection Act, 18 U.S.C. §§ 2710-12 (2000).

¹⁴ Ruan, K., Carthy, J., Kechadi, M. T., & Crosbie, M. (2011b).

¹⁵ Spyridopoulos, T., & Katos, V. (2011). Requirements for a Forensically Ready Cloud Storage Service. *International Journal of Digital Crime and Forensics*, 3(3), 19–36.

¹⁶ Taylor, M., Haggerty, J., Gresty, D., & Hegarty, R. (2010). Digital evidence in cloud computing systems. *Computer Law & Security Review*, 26(3), 304–308.

¹⁷ Taylor, M., Haggerty, J., Gresty, D., & Lamb, D. (2011). Forensic investigation of cloud computing systems. *Network Security*, 2011(3), 4–10. doi:10.1016/S1353-4858(11)70024-1.

¹⁸ Wolthusen, S. D. (2009). Overcast: Forensic Discovery in Cloud Environments (pp. 3–9). Presented at the IT Security Incident Management and IT Forensics, 2009. IMF '09. Fifth International Conference on. doi:10.1109/IMF.2009.21.

¹⁹ Yan, C. (2011). Cybercrime forensic system in cloud computing (pp. 612–615). Presented at the Image Analysis and Signal Processing (IASP), 2011 International Conference on. doi:10.1109/IASP.2011.6109117.

It is important for the reader to understand this; even if there is no legal barrier preventing a provider of cloud services from providing access to cloud-stored-data to third parties, absent some contractual obligation to the contrary the service provider can still refuse to provide access. A legally binding directive thus serves two purposes. It meets any constitutional or other barriers to disclosure and it mandates disclosure, whether there is a barrier or no.

It is also important to understand that a directive can be legally binding in that it requires the recipient to comply or face contempt or other court action, and still not be legally sufficient in terms of complying with the constitution or statute. For example, in most jurisdictions, attorneys in civil and criminal cases routinely issue "trial subpoenas duces tecum" compelling the attendance of witnesses in court with specified documents. If properly served, a witness receiving such a subpoena must appear or contest the subpoena or face possible contempt of court citation. But if this subpoena was issued for constitutionally protected documents, requiring a warrant supported by probable cause, or for documents covered by specific statutes limiting government access, the subpoena will not be sufficient to save the documents from suppression or the provider and government from civil sanction, or both.

2.2 Legal Limits on Access by Law Enforcement

Legal limits on access to cloud data by or on behalf of law enforcement flow from two sources: constitutional limits (Federal and state), and statutory limits.²⁰ These restrictions on searches have different impacts. Searches that violate constitutional protection are likely to be suppressed. Searches that violate statutory restrictions may be subject to suppression, but more typically, while providing a civil remedy to the aggrieved party, do not provide suppression as a remedy. Either way, it is wise to insure one is using a legal production mechanism. Section 2.3 will consider the Constitutional limitations in detail and Section 2.4 will discuss the Statutory Limitations.

2.3 Constitutional Limitations

The below discussion relates to the requirements of the U.S. Constitution. These requirements set a floor below which protection must not fall. The courts of several states, however, have interpreted their state constitutions to be more protective of privacy than the Fourth Amendment. Investigators in those states must be aware of the standards in the state or states in which they operate.

Constitutional limitations flow from the Fourth Amendment's prohibition against unreasonable searches and seizures.

The right of the people to be secure in their persons, houses, papers, and effects, against unreasonable searches and seizures, shall not be violated, and no Warrants shall issue, but upon probable cause, supported by Oath or affirmation, and particularly describing the place to be searched, and the persons or things to be seized." U.S. CONST. amend. IV.

There are three questions to be asked in the Fourth Amendment context: Does the Fourth Amendment apply to the situation; Is there a legal warrant or warrant equivalent; and, if there is not a legal warrant, Does the search fall within the exceptions to the warrant requirement?

2.3.1. Does the Fourth Amendment Apply to the Situation?

²⁰ Limits can also be imposed by court rule or the magistrate issuing a warrant. These are not discussed as they have only limited unique applicability to cloud stored data.

This determination requires two separate considerations:

- a. Is there a reasonable expectation of privacy?
- b. Is government action involved?

2.3.1.1. Is There a Reasonable Expectation of Privacy?

2.3.1.1.1. Reasonable Expectation of Privacy in General

One unique aspect of cloud computing is the data is stored, not at the user's home or office, but away from that location, in the hands of third parties. Another unique aspect is that the user may be viewed as having given explicit or implicit consent for others to access the data. These two factors have the potential to preclude a user from having any reasonable expectation of privacy in cloud data.

This is a new and developing area of the law. When this developing law is coupled with evolving societal and legal concepts of when one is entitled to privacy, little can be said with certainty about how courts will ultimately determine the issue of whether, and under what circumstances, there can be a reasonable expectation of privacy in cloud data.

Because this can be a confusing area of the law, we devote significant time to explanation and exploration of this topic.

In determining whether there is a reasonable expectation of privacy, courts look at two factors: the subjective expectation of the person asserting a privacy interest and whether society accepts that as objectively reasonable.²¹ The highest privacy interest is attached to private dwellings.²² A computer inside a private dwelling will certainly be accorded a similar privacy status.

Expectation of privacy in the contents of a computer is less clear when the computer is not in the home or when the contents of the computer are routinely shared with others or members of the public. Cloud computing often involves situations where a computer in the home (or workplace) connects to or accesses data stored outside the home (or workplace). Furthermore, while data in the cloud may be considered to be "shared" with the cloud provider, it is highly unlikely that cloud users intend this to mean their data is shared with the public unless the user explicitly takes such action. This hybrid of within-the-home/outside-the-home, private/shared-with-the-provider computing can create challenges to courts attempting to discern the boundaries within which one has a reasonable expectation of privacy.

To understand the law in this area, some background is needed. In *Katz v. United States*,²³ the Supreme Court expressly held that "what a person knowingly exposes to the public, even in his home or office, is not a subject of Fourth Amendment protection."²⁴ The Supreme Court "consistently has held that a person has no legitimate expectation of privacy in information he voluntarily turns over to third parties."²⁵

This general principle, that one has no reasonable expectation of privacy in information he shares with third parties, can be justified in two ways.

²¹ *California v. Greenwood*, 486 U.S. 35, 39, 108 S.Ct. 1625, 100 L.Ed.2d 30 (1988).

²² *Welsh v. Wisconsin*, 466 U.S. 740, 748, 104 S.Ct. 2091, 80 L.Ed.2d 732 (1984).

²³ 389 U.S. 347, 104 S.Ct. 2091, 80 L.Ed.2d 732 (1984).

²⁴ *Id.* at 351.

²⁵ *Smith v. Maryland*, 442 U.S. 735, 743-44, 99 S.Ct. 2577, 61 L.Ed.2d 220 (1979); *see, e.g., U.S. v. Miller*, 425 U.S. 435, 442-43, 96 S.Ct. 1619, 48 L.Ed.2d 71 (1976); *Couch v. U.S.*, 409 U.S. 322, 335-36, 93 S.Ct. 611, 34 L.Ed.2d 548 (1973).

The first, often referred to as “disclosed to the public”, deals with the manner and nature of the sharing. If an individual shares information with the public in general, whether through standing on a street corner and yelling, publishing it in publications intended for the public at large, or posting it in a public forum, there can be little disagreement that the individual has no privacy interest in the information. This meets neither test – there is no subjective expectation of privacy and society is not willing to recognize a claim of privacy under those facts.²⁶ Data disseminated in this manner, cloud stored or otherwise, is not protected under the Fourth Amendment. For example, many blogs are stored on third party servers. If those blogs are “published”, i.e., made available to the public at large, then there is no reasonable expectation of privacy.

The second justification, usually referred to as the “third-party doctrine” focuses on the risk the individual assumes, in disclosing information to a third party, that the third party will disclose the information to others, including law enforcement. This is the form of shared information that is important in the cloud context.

Prof. Orin Kerr at George Washington University Law School has argued that the “third-party doctrine” should more properly be viewed under the consent exception to the warrant requirement, rather than as a determinant of whether the defendant had a reasonable expectation of privacy.²⁷ Though Prof. Kerr’s analysis is convincing, courts have continued to view third-party involvement under the reasonable-expectation-of-privacy analysis and so shall we. We will return, however, to the consent exception to the warrant requirement later in this chapter.

2.3.1.1.2. Reasonable Expectation of Privacy in the Cloud

With that background information, the question becomes does an individual or entity have a reasonable expectation of privacy in data she has stored in the cloud? Does storing data on third party computers constitute “exposing it to the public” or “voluntarily turning it over to third parties” so as to vitiate any privacy expectation?

Although, as noted earlier, this is a two-part test (does the individual have a subjective expectation of privacy and does society recognize that expectation as reasonable), most decisions in this area are decided under the second part of the test – whether the expectation of privacy is reasonable.²⁸

The problem in answering that question is that, as the Supreme Court has noted, “We have no talisman that determines in all cases those privacy expectations that society is prepared to accept as reasonable.”²⁹

Still, scholars have attempted to discern, from court decisions, the logical framework behind these decisions,³⁰ but have largely concluded that the perceived logical framework is more useful in characterizing decisions than in providing guidance for future decisions.³¹

Nonetheless we will attempt to characterize, from case law, some factual and policy elements that make a finding of a reasonable expectation of privacy more, or less, likely.

²⁶ *California v. Greenwood*, 486 U.S. at 39.

²⁷ Orin S. Kerr, *The Case for the Third-Party Doctrine*, 107 Mich. L. Rev. 561, 588-90 (2009).

²⁸ If the answer is that the very act of storing ones data on the hard drive of another party vitiates any reasonable expectation of privacy, then no further discussion is needed because such storage-with-third-parties is at the core of cloud storage. It is likely, however, as with many things legal, that the answer will not be so absolute, and a more refined analysis is needed.

²⁹ *O'Connor v. Ortega*, 480 U.S. 709, 715, 107 S.Ct. 1492, 94 L.Ed.2d 714 (1987) (plurality).

³⁰ Wayne R. LaFave, *Search and Seizure: A Treatise on the Fourth Amendment* § 2.1(b), at n. 82.1 (4th ed. 2004 & Supp. 2008–09), and Orin S. Kerr, *Four Models of Fourth Amendment Protection*, 60 Stan. L.Rev. 503 (2007)).

³¹ See Orin S. Kerr, 107 Mich. L. Rev. 561.

2.3.1.1.2.1. Reasonable Steps Taken to Conceal the Contents.

This category includes the steps taken by the individual to minimize the likelihood that the data can be viewed by anyone other than those the individual voluntarily exposes it to. In the non-cloud world this includes using opaque containers and keeping those containers in close proximity,³² locking luggage or a briefcase,³³ or password protecting or encrypting files, directories, or a computer.³⁴

These steps go directly toward demonstrating a subjective expectation of privacy by decreasing the likelihood or probability of disclosure to others. Likewise, the use of such concealment mechanisms, such as unlisted URL, password protected access and data encryption in cloud storage are strong indications of a subjective privacy expectation.

2.3.1.1.2.2. Nature of the Contents

Courts sometime correlate a reasonable expectation of privacy to the nature of the container and its contents. For example, *U.S. v. Frierie*³⁵ held the defendant had a reasonable expectation of privacy in a briefcase he had entrusted to a third party, noting that “fewer places outside one’s home justify a greater expectation of privacy than a briefcase.”³⁶

Courts have viewed the contents of a computer as similar to the contents of a briefcase, and equally entitled to a heightened expectation of privacy.³⁷ There is reason to believe that courts will find the same heightened expectation of privacy in the same kind of data stored in the cloud.

2.3.1.1.2.3. Absence of Evidence of Abandonment of Expectation of Privacy

No person can have a reasonable expectation of privacy in property he has abandoned.³⁸ To demonstrate abandonment the Government must establish that the defendant’s voluntary words or conduct would lead a reasonable person to believe that the defendant relinquished his property interests in the item searched/seized.³⁹

Absent some specific statement by the defendant the only thing that would allow a court to consider the defendant to have abandoned her expectation of privacy in cloud data would be a sustained period of inactivity in the account.⁴⁰

³² *Bond v. United States*, 529 U.S. 334, 120 S.Ct. 1462, 146 L.Ed.2d 365 (2000), held that a bus passenger exhibited a subjective expectation of privacy in the contents of his carry-on luggage “by using an opaque bag and placing that bag directly above his seat.”

³³ *United States v. Presler*, 610 F.2d 1206, 1213-14 (4th Cir. 1979) (“The very act of locking [briefcases] and retaining either the key or the combination to the locks on the two briefcases was an effective expression of the defendant’s expectation of privacy.”)

³⁴ *United States v. Ziegler*, 474 F.3d 1184, 1189 (9th Cir. 2007) (“The government does not contest Ziegler’s claim that he had a subjective expectation of privacy in his office and the computer. The use of a password on his computer and the lock on his private office door are sufficient evidence of such expectation.”)

³⁵ 710 F.2d 1515, 1519 (11th Cir. 1983).

³⁶ *Id.*

³⁷ *United States v. Andrus*, 483 F.3d 711, 718 (10th Cir.2007), *cert. denied*, 522 U.S. 12997, 128 S.Ct 1738. 170 L.Ed.2d 542 (2008). (“A personal computer is often a repository for private information the computer’s owner does not intend to share with others . . . intimate information is commonly stored on computers, it seems natural that computers should fall into the same category as suitcases, footlockers, or other personal items that command a high degree of privacy.”)

³⁸ *Hester v. United States*, 265 U.S. 57, 58, 44 S.Ct. 445, 68 L.Ed 898 (1924).

³⁹ *United States v. Stephens*, 206 F.3rd 914, 917 (9th Cir. 2000).

⁴⁰ However, even a prolonged period of inactivity might not be indicative of abandonment where the cloud is being used solely as a static backup of fixed data like heirloom photographs or video.

2.3.1.1.2.4. Were the Documents in Question Created by the User or Were they Created by the Third-Party

The Supreme Court case in which they found no constitutional expectation of privacy in bank records, *United States v. Miller*,⁴¹ dealt with the bank's records, specifically the business records created or by the bank that detailed transactions with the defendant. That is, the defendant took certain actions regarding his bank account—a withdrawal or deposit for example—and the bank created records of these transactions. Sometimes the record is one totally created by the bank, such as monthly account statements. Other records maintained by the bank are copies of items created by the user, such as checks. In all instances, however, the records actually disclosed were copies made from the microfiche copies the bank maintained. *United States v. Graham*⁴² applied the same logic, that the records disclosed were the third party's records, not the individual's records, to cell site location records.

This “who-created-the-records” standard is not always followed. For example, the Supreme Court found no reasonable expectation of privacy in records the taxpayer provided to an accountant, even though these were original records of the taxpayer.⁴³

To the extent who created the records determines whether there is a reasonable expectation of privacy, in the cloud context the actual files were created by the user, not the cloud provider. If the records sought consist, instead, of transactional records, such as log files of cloud access, created by the cloud provider, it is less likely that courts will find the user to have a constitutional privacy interest in those records.

2.3.1.1.2.5. Whether the Nature of the Relationship between the Individual and the Third-Party Requires the Third-Party to have Access to the Contents Provided

An easy example of a relationship that requires access to the contents of the information provided is the banking relationship. When you write a check that is ultimately presented to your bank, the bank does more than note that a check has been written against your account. Numerous substantive content items are examined by the bank including the account and check number imprinted on the check, the amount, the signature, and in many instances, the payee and endorsement.

A contrary example is the telephone company who, for legitimate business reasons, needs to know what numbers you've called, and when, but ordinarily has no legitimate reason to monitor the content of your calls.

This approach in the cloud context—providing protection to transmission to/through third parties where the relationship only requires the third party to examine “routing information” (the electronic equivalent of the address on the outside of an envelope) but not finding a reasonable expectation of privacy in relationships where the third party is expected to or entitled to examine the content of the relationship—has been advocated by Professor Kerr,⁴⁴ but has not always been given a warm reception by courts.⁴⁵

⁴¹ *United States v. Miller*, 425 U.S. 435.

⁴² CRIM. RDB-11-0094, 2012 WL 691531 (D. Md. Mar. 1, 2012) (“Like the bank records at issue in *Miller*, the historical cell site location records in this case are not the “private papers” of the Defendants—instead, they are the “business records” of the cellular providers.”)

⁴³ *Couch v. U.S.*, 409 U.S. 322

⁴⁴ Orin S. Kerr, *Applying the Fourth Amendment to the Internet: A General Approach*, 62 Stan. L.Rev. 1005, 1038 (2010) (“[T]he key point is that the third-party doctrine has not been extended to intermediaries that merely send and receive contents without needing to access or analyze those communications. Instead,

One case discussed the related issue of privacy expectation in stored email where the service provider had a contractual right of limited access. In *Warshak v. United States*⁴⁶ the government obtained the defendant Warshak's stored emails by subpoena (after first requesting NuVox, the ISP holding the emails, to preserve emails in storage and prospectively, after receiving the subpoena.) Warshak sought to suppress these emails as having been obtained in violation of the Fourth Amendment. The court analyzed the issue as we have, first determining that Warshak had a subjective expectation of privacy in his email, and then examining whether society was willing to recognize this expectation as legitimate. In analyzing this latter issue the court examined the protection afforded other forms of communication that pass through intermediaries (telephone conversations and letters) and concluded that it would defy common sense to give emails less protection than letters and phone calls.

The court then addressed the government's contention that is relevant to our discussion. The government argued that the terms of service between Warshak and NuVox allowed NuVox to access Warshak's emails for some purposes.⁴⁷ The court dispensed with the government's argument as follows:

First, the court noted that the mere *ability* of a third party to access contents of a communication was not sufficient to extinguish a privacy right.

Second, and more important, the court noted that a *right* of access also did not extinguish an otherwise existing privacy right. The right of access referenced was the same as was available to many telephone companies, the right to access and use information in the operation of the service and as necessary to protect the service. The degree of this access did not operate to extinguish the privacy right in the stored email. The court likened this agreed access to that granted by hotel guests to cleaning maids, and apartment tenants to repairpersons, noting that neither of those types of access vitiated any existing privacy interest.

Finally, the court commented that their holding did *not* mean that a subscriber agreement could *never* be broad enough to snuff out a privacy right, giving as an example a reserved right by the provider to "audit, inspect, and monitor" a subscriber's emails being sufficient access to eliminate any privacy right.

The court found the government's access to Warshak's emails without a warrant to be a Fourth Amendment violation, but allowed the evidence to be used under the "good faith" exception.⁴⁸

This decision makes it clear that the exact nature of the relationship between the provider and user, and any agreed access the user provides in the service agreement, can have a major impact on whether a reasonable expectation of privacy will be recognized.

2.3.1.1.2.6. The Contractual Relationship between the Individual and the Third-Party

courts have widely adopted the content/non-content line or a functional equivalent in cases applying the Fourth Amendment to communications networks.” (emphasis added))

⁴⁵ *In re U.S. for an Order Authorizing the Release of Historical Cell-Site Info.*, 809 F. Supp. 2d 113, 125 (E.D.N.Y. 2011) (“[T]he “without needing to access or analyze those communications” caveat appears too narrow to accommodate current case law and the realities of current technology. For example, would the third-party doctrine remove the reasonable expectation of privacy over the contents of emails sent on Gmail, or similar email providers, that use computers to access and analyze the contents of email communications in order to display advertisements? While the court need not answer the question here, it appears that such a technology-specific definition of third-party disclosure would fail to take into account the importance of the information disclosed or the intrusive nature of disclosing such information.”)

⁴⁶ *Warshak v. United States*, 631 F.3d 266 (6th Cir. 2010).

⁴⁷ *Warshak*, 631 F.3d at 286

⁴⁸ *Illinois v. Krull*, 480 U.S. 340, 348, 107 S.Ct. 1160, 94 L.Ed.2d 364 (1987), citing *United States v. Leon*, 468 U.S. 897, 916, 104 S.Ct. 3405, 82 L.Ed.2d. 677 (1984)

A closely related question is whether the contract or service agreement provides that the third-party may “mine” the content stored by the third-party? If so, the user is tacitly providing the third party not only custody of the data but also access permission. While such permission may not be conclusive on the issue of whether a reasonable expectation of privacy exists, the absence of any such content permission argues in favor of a finding of a protected privacy interest.

2.3.1.1.3 Conclusion as to Reasonable Expectation of Privacy in Cloud Data

As noted, cloud storage creates some fairly unique issues regarding reasonable expectation of privacy because of the shared or quasi-shared nature of the storage. Resolution of this issue—whether there is a reasonable expectation of privacy in data stored in the cloud—will ultimately be made by the courts, and the trend appears to be in the direction of finding a protected privacy interest.⁴⁹ Two respected commentators have urged the courts to hold that such protection exists.⁵⁰ In the interim the prudent investigator will proceed on the assumption that privacy rights exist, and will only proceed with a warrant or warrant equivalent or under one of the exceptions to the warrant requirement. When this reasonable-expectation-of-privacy question is viewed as a consent issue, however, the answer may show this privacy interest to be useless if the cloud provider can consent to a search (as discussed in 2.7.1 below).

2.3.1.2. Is There Government Action?

The Fourth Amendment only applies to the actions of the government and third parties acting as agents of the government. The exclusionary rule was not intended and does not apply to the actions of private citizens or foreign law enforcement agents (including law enforcement of another state). The government may use information gathered by private citizens and foreign law enforcement under the “silver platter” doctrine.

2.3.1.2.1. The Silver Platter Doctrine

The silver platter doctrine arose in the early 1900s when federal search and seizure law was more restrictive than state search and seizure law, and suppression was a remedy for violation of the federal law. When state law enforcement officers conducted a search legal under state law but illegal under federal law, could the fruits of the search be used in federal courts? The answer was yes, if the federal authorities received the fruits on a silver platter, i.e., they were not so involved in the illegal search as to require suppression.

The “silver platter” doctrine withered on the vine after the protections of the Bill of Rights were extended to the states in *Wolf v. Colorado* (1949)⁵¹ and state and Federal protections were equal; however, starting in the 1980s, some state courts began holding their own constitutions to be more protective than the Federal constitution and the flow of evidence started moving the other way. Now Federal law enforcement (or law enforcement of other states), facing less stringent legal requirements, started providing evidence to state police on this same silver platter.⁵² At best,

⁴⁹ See *Warshak v. United States*, 490 F.3d 455, 470 (6th Cir. 2007), *vacated* 532 F.3d 521 (6th Cir. 2008) (the provisions of the decision relevant to the text were reinstated in 631 F.3d 266 (6th Cir. 2010)), *Quon v. Arch Wireless Operating Co., Inc.*, 529 F.3d 892 (9th Cir. 2008) *rev'd and remanded sub nom. City of Ontario, Cal. v. Quon*, 130 S. Ct. 2619, 177 L. Ed. 2d 216 (U.S. 2010), and *State v. Bellar*, 231 Or. App. 80, 217 P.3d 1094 (2009). Although the privacy issue was not reached by the Supreme Court in *Quon*, and the relevant discussion in *Bellar* was in a dissent, the logic of the decisions remain.

⁵⁰ Kerr at footnote 45, *supra*, and David A. Couillard, *Defogging the Cloud: Applying Fourth Amendment Principles to Evolving Privacy Expectations in Cloud Computing*, 93 MINN. L. REV. 2205, 2216 (2009).

⁵¹ 338 U.S. 25, 69 S.Ct. 1359, 93 L.Ed. 1782 (1949).

⁵² See *N.J. v. Mollica*, 554 A.2d 1315, 1324 (N.J. 1989) for a superb discussion of the history of the silver platter doctrine.

of course, this only means that the higher state standard may be avoided. Compliance with the Federal standard as a floor is always required.

Courts view a transfer from a Federal official to a state official as analogous to a transfer from a private citizen to a state official. Searches by private citizens, not acting as the agent of law enforcement, are not entitled to constitutional protection.

2.3.1.2.2 Conclusion as to Government Action and the Cloud

The standard will almost certainly be the same for cloud data as for local data. The primary difference is that the nature of the relationship and the service agreement between the user and the cloud provider may create more situations where the provider will monitor, and even examine, cloud data for contraband. For example, under a Federal statute, since repealed,⁵³ AOL monitored communications for child pornography and reported any such material to a tip line which then reported the tip to law enforcement. This was found not to violate any Fourth Amendment right of the defendant because the scanning was done by a private entity to recognize files possibly detrimental to their system.⁵⁴ Similar logic would likely apply to other monitoring done by cloud providers, such as monitoring for material violating copyright or to data mining done under the service agreement.

2.3.1.3 Conclusion re Applicability of the Fourth Amendment to Cloud Data

If either of these elements is missing—reasonable expectation of privacy or government action—the Fourth Amendment does not apply. If the statutory protections described below do not apply then there is no legal requirement to obtain a warrant, court order or subpoena.⁵⁵

If both of these elements are present then the investigator must have either a warrant (or warrant equivalent) or fall within one of the consent exceptions described below.

2.4 Statutory Limitations - Electronic Communications Privacy Act

In response to decisions like *Miller*,⁵⁶ finding no expectation of privacy in bank records, Congress has passed a number of laws to provide statutory privacy protection for bank records,⁵⁷ cable television viewing and video rentals.⁵⁸ Congress also extended the wiretap statute to provide some privacy protection to records related to electronic communications in the Electronic Communications Privacy Act.⁵⁹ Thus, at the Federal level, in many situations involving third party record holders, any privacy protection in third parties records stems from statutes, not the constitution.

The most relevant of these to our discussion is the Electronic Communications Privacy Act.

⁵³ 42 U.S.C. § 13032(b)(1), since recodified in 18 U.S.C. § 2258A.

⁵⁴ *United States v. Richardson*, 607 F.3d 357, 363 (4th Cir. 2010) cert. denied, 131 S. Ct. 427, 178 L. Ed. 2d 324 (U.S. 2010).

⁵⁵ As noted earlier, the absence of constitutional or statutory protections does not mean the cloud provider *must* provide any requested data. Absent voluntary compliance, enforceable directives would still be needed to compel production.

⁵⁶ *U.S. v. Miller*, 425 U.S. 435.

⁵⁷ Right to Financial Privacy Act, 12 U.S.C. §§ 3401-22 (2000).

⁵⁸ Cable Privacy Act, 47 U.S.C. § 551 (2000) and the Video Privacy Protection Act, 18 U.S.C. §§ 2710-12 (2000).

⁵⁹ 18 U.S.C. § 2701-09 (2000).

The Electronic Communications Privacy Act ["ECPA"] updated the wiretap statute⁶⁰ and added the Stored Electronic Communications Act ["SCA"].⁶¹ Although it is common to reference ECPA in discussing limitations on computer searches, most often it is the specific provisions of SCA that are in play.⁶²

2.4.1. Provisions of the SCA

2.4.1.1. What Does the SCA Cover?

The act covers the contents of electronic communications while in electronic storage by an "electronic communications service."⁶³ An electronic communications service is "any service which provides to the users thereof the ability to send or receive wire or electronic communications."⁶⁴ The act also covers the contents of electronic communications in a "remote computing service."⁶⁵ A remote computing service is one that provides "to the public ... computer storage or processing service by means of an electronic communications system."⁶⁶

2.4.1.2. What Does the SCA Prohibit?

The act prohibits providers of electronic communication service or remote computing service from disclosing contents of electronic communications or customer records and transactional data to the government, except as provided.

2.4.1.3. When Is a Disclosure to the Government Allowed?

This complex topic is discussed in great depth in many readily available sources.⁶⁷ A discussion of this topic in similar depth is beyond the scope of this chapter. As a general rule of thumb, however, all records, except the content and record of real-time communications, can be obtained by a warrant supported by probable cause. Many, but not all records can be obtained by subpoena or court order.

2.4.1.3.1. Subscriber or Customer Records

2.4.1.3.1.1. Records Obtainable by Subpoena

Subscriber or customer records (name, address, telephone toll billing records, telephone number or other subscriber number or identity, and length and type of service for a subscriber or customer) may be obtained by subpoena (or higher process like a warrant or court order.)

The USA PATRIOT Act⁶⁸ made other subscriber/customer records, such as transaction history, IP logs, etc. available by subpoena. Previously they had been available only by warrant or court order.

⁶⁰ 18 U.S.C. §§ 2510-22 (2000).

⁶¹ 18 U.S.C.A. §§ 2701-12 (2004).

⁶² Keystroke monitoring and other real-time monitoring of electronic communications is covered by the wiretap statute, 18 U.S.C. § 2511(1) (2000).

⁶³ Id. § 2510(15).

⁶⁴ Id.

⁶⁵ 18 U.S.C.A. § 2703(b) (2004).

⁶⁶ Id. § 2711(2).

⁶⁷ See, e.g., *Searching and Seizing Computers and Obtaining Electronic Evidence in Criminal Investigations*, Office of Legal Education Executive Office for United States Attorneys (2009) pp 127-138 (hereafter "Searching and Seizing Computers").

⁶⁸ Pub. L. No. 107-56, 115 Stat. 272 (2001).

Under the PATRIOT Act, a subpoena can be used to obtain basic subscriber information including customer's name; address; length of service; means and source of payment (including any credit card or bank account number); local and long-distance telephone toll billing records; and records of session times and durations, as well as any temporarily assigned network address.⁶⁹

Other sections of the SCA, not discussed here, allow access to opened email under a subpoena with notice to the customer. Provision is made for delayed notice in some circumstances.⁷⁰

2.4.1.3.1.2. Records Obtainable by Court Order

Under the SCA, a court order, sometimes referred to as an "articulable facts order" or "§ 2703(d) court order," may be sought to obtain all other subscriber information except the content of an unopened e-mail that has been stored for 180 days or less.⁷¹ To obtain a "§ 2703(d) court order," there must be specific and articulable facts showing that there are reasonable grounds to believe that the specified records are relevant and material to an ongoing criminal investigation. These records would include complete audit trails/logs, Web sites visited, identities of e-mail correspondents, cell site data from cellular/PCS carriers, and opened e-mail. As a practical matter, a "§ 2703(d) court order" can also be used to obtain basic subscriber information. Notice to the subscriber is only required when opened e-mail is requested from the provider.⁷²

2.4.1.3.2. Contents of Electronic Communications

In general, the contents of an electronic communication may be disclosed with the consent of any party to the communication. The consent of all of the parties to the communication is not needed—only one party need consent.⁷³

A government entity can compel the production of the contents of stored electronic communications by warrant. As previously noted, however, communications in storage for more than 180 days can be obtained by subpoena.

2.4.1.4. What Is the Remedy for Violation of this Act?

2.4.1.4.1. Criminal Liability

The SCA provides for criminal liability for one who "(1) intentionally accesses without authorization a facility through which an electronic communication service is provided; or (2) intentionally exceeds an authorization to access that facility; and thereby obtains, alters, or prevents authorized access to a wire or electronic communication while it is in electronic storage in such system."⁷⁴ Punishment can be up to two years imprisonment.

2.4.1.4.2. Civil Liability

The act creates a civil cause of action for violations, including damages and attorney's fees with punitive damages if the violation was willful. Although one early decision held that only the cloud provider, not the government, was civilly liable for violations;⁷⁵ a later decision found

⁶⁹ 18 U.S.C.A. § 2703(b) (2004).

⁷⁰ *Id.* § 2703(b).

⁷¹ The initial Warshak Circuit opinion held that all unopened email, regardless of the length of time it had been held by the cloud provider, requires a warrant. That opinion was later vacated because the issue was not ripe for consideration. The portions of the decision relating to the necessity for a warrant was reinstated in the substantive appeal of the case. See footnote 50 *supra* for citations.

⁷² *See Id.* § 2703(b).

⁷³ *See* 18 U.S.C.A. § 2511(c) (2004).

⁷⁴ 18 U.S.C.A. §§ 2701(a)(1) & (2) (2004).

⁷⁵ *Tucker v. Waddell*, 83 F.3d 688 (4th Cir. 1996).

liability in the government official who obtained the information.⁷⁶ Changes in the SCA made in 2001 reconciled the provisions that led to these discrepant results, and courts have more routinely found exposure to civil liability for the government.⁷⁷

2.4.1.4.3. Suppression

Suppression is not a remedy for non-constitutional violations of SCA.⁷⁸ This protection, as noted, does not preclude suppression for constitutional violations under state constitutions. As noted earlier, in the Federal arena, the Sixth Circuit, in a since-vacated opinion, held that contrary to *U.S. v. Miller*,⁷⁹ an individual had a reasonable expectation of privacy in the content of his emails notwithstanding the fact that a third-party provider of Internet communication services had access to those messages.⁸⁰ If this decision is embraced by the other circuits or the Supreme Court, some provisions of the SCA may themselves be unconstitutional.

2.4.1.4.4. Good Faith

Section 2707(e) of the SCA states that “reliance on ... a court warrant or order, a grand jury subpoena, a legislative authorization, or a statutory authorization ... is a complete defense to any civil or criminal action brought under this chapter.”⁸¹ The Tenth Circuit used this provision in *Davis v. Gracey*⁸² to reject the defendants’ civil claim under SCA. The court noted that because the officers relied on a warrant supported by probable cause they were entitled to a “good faith” defense.⁸³ A number of states do not incorporate the good-faith defense into their own constitution.⁸⁴ It is not known what impact that doctrine will have on the good-faith provisions of the SCA in the context of civil or criminal liability.

2.4.2. Does the SCA Apply to Cloud Stored Data?

Other than the evolving interpretations of the constitutional reasonable expectation of privacy issue described earlier, the real question for this chapter is whether the SCA applies at all to cloud computing.

There are two issues that determine the answer to that question:

- Does the SCA’s limitations on access to records of electronic communications imply limitations on access to information stored in the cloud; and
- Are cloud providers included within the definitions of “electronic communication service” or “remote computing service”?

2.4.2.1. Does the SCA’s limitations on access to records of electronic communications imply limitations on access to information stored in the cloud?

The SCA defines “electronic communication” as any transfer of signs, signals, writing, images, sounds, data, or intelligence of any nature transmitted in whole or in part by a wire, radio,

⁷⁶ *McVeigh v. Cohen*, 983 F. Supp. 215 (D.D.C. 1998).

⁷⁷ *Freedman v. America Online, Inc.*, 303 F.Supp.2d 121 (D.Conn.,2004)

⁷⁸ 18 U.S.C.A. § 2708 (2004).

⁷⁹ 425 U.S. 435.

⁸⁰ *United States v. Warshak*, 631 F.3d 266 (6th Cir. 2010).

⁸¹ *Id.* § 2707(e).

⁸² 111 F.3d 1472 (10th Cir. 1997).

⁸³ *Id.* at 1484.

⁸⁴ *See State v. Afana*, 233 P.3d 879 (Wash.,2010)

electromagnetic, photoelectronic or photooptical system that affects interstate or foreign commerce.⁸⁵

Since data is transferred to the cloud electronically, does this definition cover all data transferred to the cloud or does it only cover electronic communications (primarily email) stored in the cloud. There are no known court decisions on this issue.

2.4.2.2. Are cloud providers included within the definitions of “electronic communication service” or “remote computing service”?

“Electronic Communication Service” means any service which provides to users thereof the ability to send or receive wire or electronic communications.⁸⁶ Online email cloud providers like Gmail and Hotmail appear to fall within that definition. Other cloud providers, providing only storage, would appear not to be within the definition.

“Remote Computing Service” means the provision to the public of computer storage or processing services by means of an electronic communications system.⁸⁷

Using just this language it would appear that cloud providers are providing remote computing services. The SCA is, however, extraordinarily complex. The section of the statute that limits disclosure to the government, only applies that limit to data that is maintained by the provider “solely for the purpose of providing storage or computer processing services to such subscriber or customer, if the provider is not authorized to access the contents of any such communications for purposes of providing any services other than storage or computer processing.”⁸⁸

One commentator has concluded that:

The applicability of the RCS provisions in the Stored Communications Act will, therefore, always require examining the cloud provider’s terms of service agreement and privacy policy. If a customer authorizes access to her data for the provision of contextual or targeted advertising services, the Act will not apply and the data will be at risk of disclosure to the government or another third party.⁸⁹

2.4.3. Conclusion as to Applicability of the SCA to Cloud Data

Except for cloud providers of email services, and cloud providers who have no contractual right to access the contents of cloud data, the provisions of the SCA appear not to apply to cloud data. Making such a determination, however, requires familiarity with the details of the service agreement for any service whose data one wishes to access.

If one concludes that Fourth Amendment protections apply to cloud data then a warrant is required anyway, so the question of applicability of the SCA may be moot. But Fourth Amendment law includes exceptions to the warrant requirement, described below. If an investigator is acting without a warrant under one of these exceptions, he is walking on firm constitutional ground but may still be violating the SCA.

⁸⁵ 18 U.S.C.A. § 2510 (West)

⁸⁶ *Id.*

⁸⁷ 18 U.S.C.A. § 2711 (West)

⁸⁸ 18 U.S.C.A. § 2702 (West)

⁸⁹ William Jeremy Robison. *Free at What Cost?: Cloud Computing Privacy Under the Stored Communications Act*, 98 Georgetown Law Journal 1195, 1213.

2.5. What Qualifies as a Legal Mandatory Directive?

2.5.1 When May a Subpoena be Used to Obtain Constitutionally Protected Information

Where the Constitution or statutes require a warrant, the warrant must meet constitutional standards described below. However, there may be circumstances where, because the data that is sought is in the possession of third parties, a subpoena may be used. Grand jury subpoenas and administrative subpoenas are permitted to reach areas protected by the Fourth Amendment under a variety of theories related to the historic role grand juries have played in criminal investigations.⁹⁰ The Supreme Court standard for such subpoenas is “reasonableness”.⁹¹ Some state courts have permitted a grand jury subpoena to reach things protected by the Fourth Amendment so long as the subpoena is supported by probable cause.⁹² Other courts have implied that so long as a subpoena gives prior notice and an opportunity to object a subpoena can reach otherwise protected area of privacy.⁹³

Courts have often excused a subpoena from some of the more stringent requirements of a warrant because of the different ways evidence is obtained under the two. Under a warrant the evidence is *seized*. Under a subpoena the evidence is *demand*ed. Courts have stated that use of a subpoena supported by probable cause often provides procedural safeguards that even a warrant does not provide.⁹⁴

The importance of this distinction flows from the fundamental difference between cloud forensics and non-cloud forensics. While in non-cloud forensic situations the investigator will be working with the actual hard drive(s) at issue which have been seized under a warrant, mandatory disclosure processes to obtain cloud data, whether warrant or subpoena in form, will, except perhaps in the most severe national security context, always be executed as if they were a subpoena, i.e., a document will be served on the record holder and that record holder will examine her files and produce the responsive data. Several courts have noted that warrants, supported by probable cause, but executed as subpoenas, where the civilian record holder actually conducts the search, provide more privacy protections.⁹⁵

⁹⁰ “[R]ecognizing the historical function of the grand jury as an independent investigatory body which acts as a buffer between the citizenry and government, the Court has been reluctant to measure the enforceability of grand jury subpoenas against the same standards applicable to search warrants. *See, e.g., Hale v. Henkel*, 201 U.S. 43, 76, 26 S.Ct. 370, 50 L.Ed. 652 (1906) (holding that while a grand jury subpoena for the production of books and papers may implicate the Fourth Amendment, when presented with a challenge to such a subpoena a court need only determine whether the request is “far too sweeping in its terms to be regarded as reasonable.”); *United States v. Calandra*, 414 U.S. 338, 346, 94 S.Ct. 613, 38 L.Ed.2d 561 (1974) (quoting *Henkel* and again suggesting that Fourth Amendment rights are adequately protected when grand jury subpoenas are subjected to court review for reasonableness, but not for “probable cause”).” *In re Grand Jury Proceedings Involving Vickers*, 38 F. Supp. 2d 159, 162 (D.N.H. 1998)

⁹¹ *Id.*

⁹² *People v. Watson*, 214 Ill. 2d 271, 825 N.E.2d 257 (2005)

⁹³ *State v. Miles*, 160 Wash. 2d 236, 156 P.3d 864 (2007)

⁹⁴ *People v. Mason*, 989 P.2d 757, 761 (Colo. 1999)

⁹⁵ *United States v. Bach*, 310 F.3d 1063, 1067 (8th Cir. 2002) (“Civilian searches are sometimes more reasonable than searches by officers. *Harris v. State*, 260 Ga. 860, 401 S.E.2d 263, 266 (1991) (stating that a dentist may execute a search warrant for dental X-rays and impressions); *Schalk v. State*, 767 S.W.2d 441, 454 (Tex.App.1988) (providing that a search by a civilian software expert more reasonable than search by an officer because the officer lacked knowledge to differentiate a trade secret from a legitimate computer software program), cert. denied, 503 U.S. 1006, 112 S.Ct. 1763, 118 L.Ed.2d 425 (1992); *State v. Kern*, 81 Wash.App. 308, 914 P.2d 114, 117–18 (1996) (indicating that it is reasonable to delegate search

It remains to be seen whether this concept will actually result in warrants-executed-like-subpoenas being held to a lower standard.

2.6. What Constitutes a Valid Warrant?

As with other portions of this paper, this section will first describe, in general terms, the requirements for a valid warrant and then highlight any special issues involved in a warrant for cloud data.

2.6.1. The Application for the Warrant Must Demonstrate Probable Cause to Believe That Evidence of a Crime Will Be Found in the Place to Be Searched

If there is probable cause to believe *both* that a crime has occurred *and* that evidence of the crime will be found at a particular location, then a search warrant may be obtained.

2.6.1.1. Evidence of a Crime

Establishing probable cause that evidence of a crime will be found is usually a non-issue. The nature of the investigation, the nature of the crime and the nature of the evidence sought usually provide a sufficient basis for making this showing. Even though usually easy to establish, courts have made it clear that failing to establish a nexus between the crime and the items to be searched will result in suppression.⁹⁶

There will likely be no unique problems associated with cloud storage with regard to establishing this nexus.

2.6.1.2. Will Be Found in the Place to Be Searched

This is the most commonly overlooked element in search warrants: ‘What is the probable cause for believing that evidence of the crime you are investigating will be found in the location you wish to search?’ This requires much more specificity than some generic “where else would it be” explanation. Articulable facts pointing to the location are required.

This is often problematic in any application seeking to search a computer for, at the very least, you must establish probable cause to believe that the evidence you are seeking will be found in a particular residence.

It could prove to be especially problematic in the cloud context. The application for the warrant must point to specific facts demonstrating the likelihood that the data being sought will be found not on the defendant’s computer(s) in general, but specifically at the cloud location the application seeks to search. This topic is discussed in more detail in section 2.4.3 below.

of bank records to bank employees, even when police officer was not present during the search). Civilian searches outside the presence of police may also increase the amount of privacy retained by the individual during the search. *See Rodrigues v. Furtado*, 410 Mass. 878, 575 N.E.2d 1124 (1991) (body cavity search done outside presence of officers); *Commonwealth v. Sbordone*, 424 Mass. 802, 678 N.E.2d 1184, 1190, n. 11 (1997).

⁹⁶ In *State v. Nordlund*, 53 P.3d 520 (Wash. App. 2002), *petition for review denied*, 70 P.3d 964 (Wash. 2003), the police were investigating several instances of attacks against young girls and sought to search the suspect’s computer. The pertinent part of the affidavit said: “[Nordlund] used a computer at this residence to access pornography and to communicate with others via E-mail. Therefore, the computer and any electronic storage media could likely be important evidence in this case regarding intent, dates and locations.” *Id.* at 525. The court noted there was no nexus between the alleged crimes and Nordlund’s use of the computer to access pornography and send e-mails. “Rather, it appears that the State was fishing for some incriminating document which is precisely what the first and fourth amendment prohibit.” *Id.*

2.6.2. Signed by a Neutral and Detached Magistrate

The warrant must be signed by a magistrate who has authority to issue the warrant. Obviously, the judge must occupy a position that has been properly created and has authority to issue warrants.⁹⁷

Special problems can arise in the context of searches of locations outside the issuing court's jurisdiction. This will almost always be the case in searches for data stored in the cloud. The judge must have authority to authorize a search of the location covered by the warrant. Statutes and court rules in some states expressly limit a judge's search warrant authority to within that state. A warrant issued by a judge from one of those states, to search a cloud provider in another state is beyond that judge's authority. If the defendant had a recognizable privacy interest in the evidence provided under the warrant, the evidence is subject to suppression.

In other states, it is less clear that there are such jurisdictional limits on warrants. For example, where a warrant issued under probable cause is being executed like a subpoena (i.e., no "search" takes place—the warrant is served on the third party who then provides the evidence requested by the warrant), it is not clear that jurisdictional limits make such a warrant invalid.⁹⁸

Several states, including California, Florida, Minnesota and Washington, have laws requiring entities doing business in their state to accept that state's process for their business records, no matter where stored. These laws also generally provide that entities incorporated in that state must accept process from other states.⁹⁹

This process has at least two limitations. First, as noted, only a small handful of states have such laws. Second, while the law purports to give judges in those states jurisdiction to *issue* such warrants or other compulsory process, it is not clear that they establish a valid mechanism for *enforcement* of the warrants and other process. What can the issuing court do if the party served with an order of production (whether in the form of a warrant supported by probable cause or other procedure) refuses to comply, or more specifically, simply ignores the order. The law of the extra territorial impact of warrants and other orders has only rarely been litigated,¹⁰⁰ and it is simply unknown whether such an out of state order, even if properly issued, can be enforced.

There are other options as well. One possibility is for law enforcement in the jurisdiction desiring the data to provide an affidavit to law enforcement in the jurisdiction in which the data is located. The data location law enforcement will obtain a local warrant for the data, incorporating the referring state's affidavit. This is commonly referred to as "domesticating" a warrant.

⁹⁷In Washington State, a district court commissioner was found to lack the authority to issue search warrants when the office of district court commissioner was not properly created. *State v. Moore*, 871 P.2d 1086 (Wash. App. 1994).

⁹⁸The Eighth Circuit addressed this issue in a circuitous way in the *Bach* case discussed in footnote 96, *supra*. The Court reversed a decision of the trial court suppressing evidence obtained from Yahoo by a search warrant issued by a state court in Minnesota. The warrant was faxed to Yahoo who complied by sending the requested information to the Minnesota authorities.

⁹⁹See, e.g., CAL. PENAL CODE § 1524.2 (West 2000) (requiring out-of-state corporations qualified to do business in California to accept service of, and comply with, California search warrants). Washington, Minnesota and Florida have similar provisions.

¹⁰⁰Orin Kerr, *Does the Fourth Amendment Allow Extraterritorial State Search Warrants*, <http://www.volokh.com/2010/01/08/does-the-fourth-amendment-allow-extraterritorial-state-search-warrants/> last viewed on August 12, 2012.

A second option is use of the Uniform Act to Secure the Attendance of Witnesses from Without a State in Criminal Proceedings.¹⁰¹ This law, a version of which exists in every state, creates procedures under which a party to a criminal case in one State can obtain the presence of a witness located in another State. This can include compelling a witness to produce evidence located in another State. The Supreme Court has found this process to be constitutional.¹⁰²

Each of these last two options can be enforced. But none of the options give state courts authority over data located outside the United States. Obtaining such data requires reference to treaties, if any, between the United States and the foreign jurisdiction and requires the involvement of the Department of Justice.¹⁰³

Of course, federal law enforcement has access to judicial districts covering the entire United States. While cumbersome, it is possible for federal law enforcement to obtain warrants in each of the districts in which the sought-for data might be found.¹⁰⁴ Federal law enforcement has the same limitations and options regarding out-of-country data.

2.6.3. Reasonably Precise in Describing the Place to Be Searched and the Items to Be Seized

There appear to be no particularity problems regarding the "items to be searched" that are unique to cloud searches. The problem relates to limits on describing the "place to be searched." While in general this should not be more burdensome for cloud searches than for other searches provided the warrant limits itself to searching for data within the target's area of the cloud, the varying and usually proprietary nature of storage and retrieval systems may present problems. Since such an area is likely demarked logically rather than physically, proof of the precision of the search may require presenting some information to the magistrate about the nature of file allocation utilized by the target cloud provider. Cloud providers may be hesitant to provide such information except under a protective order.

The precise location of cloud stored information may not be known at crucial early stages of an investigation. This could be a major stumbling block.

Even more onerous, defining "the place to be searched" in a cloud context raises many questions. Does "the place" mean the location of the business or registered agent's office for the cloud provider? Does it mean the location of "central server", if there be such a beast, for the cloud provider? Does "the place" mean the physical location of every storage device containing responsive data?

In the above section we discussed how the requirement that a warrant be signed by a neutral and detached magistrate raised the issue of jurisdiction. What is at issue here is a related, but different, location issue.

¹⁰¹“UNIFORM ACT TO SECURE THE ATTENDANCE OF WITNESSES FROM WITHOUT A STATE IN CRIMINAL PROCEEDINGS.” Paper drafted at the 41st Annual Conference of the National Conference of Commissioners on Uniform State Laws. Atlantic City, NJ. (1936) https://www.law.upenn.edu/library/archives/ulc/fnact99/1920_69/uasaw36.pdf.

¹⁰² *New York v O'Neill*, 359 US 1, 79 S.Ct. 564, 3 L.Ed.2d 585 (1959)

¹⁰³See generally “Treaty Requests” in *United States Attorneys’ Manual*, last viewed August 12, 2012, http://www.justice.gov/usao/eousa/foia_reading_room/usam/title9/crm00276.htm.

¹⁰⁴ See *Searching and Seizing Computers and Obtaining Electronic Evidence in Criminal Investigations*, Office of Legal Education Executive Office for United States Attorneys (2009) at pp. 84-86 for examples of advice given for searches of computers located in multiple jurisdictions, including out-of-country locations.

If the investigator cannot tell the warrant judge where he wants to search because he doesn't know where the data actually is stored, can he obtain a valid warrant?

This is a crucial, but rarely analyzed problem. Even the 2009 version of the comprehensive and highly regarded Department of Justice's Computer Search Manual¹⁰⁵ does not discuss the constitutional problems associated with specifying the search location in warrants for cloud based data.¹⁰⁶

The Fourth Amendment requires that warrants "particularly describ[e] the place to be searched."¹⁰⁷ The precision required is determined by whether the warrant describes the place to be searched with "sufficient particularity to enable law enforcement officers to locate and identify the premises with reasonable effort" and "whether any reasonable probability exists that the officers may mistakenly search another premise."¹⁰⁸

The kinds of cases where an absence of absolute precision has been excused include where the house is specifically and uniquely described but the wrong address is given, or other situations where sufficient specific information was provided to the searching officer so as to minimize the possible of a mistaken search of another premise.

On the other hand, finding a lack of particularity in describing the place to be searched, courts have declined validity to, and suppressed the results of, a warrant that did not specify a location other than within the issuing judge's jurisdiction - the Western District of Wisconsin.¹⁰⁹ Courts have made it clear that a warrant for a specific item at a specific location is limited to a search of that location. Officers may not use this warrant to seize the item after it was moved to a different location without a new warrant.¹¹⁰

This legal requirement may prove to be a fatal stumbling block in seeking a warrant where the location of the data in the cloud is unknown or cannot be determined.

One possible solution flows from the language of the state statutes, discussed above, authorizing state judges to compel the production of evidence, regardless of where that evidence is located. The premise of these statutes is that any entity doing business in a state is generally required to have designated a "registered agent" in the state to accept process. The process, served as a subpoena even if it is a warrant supported by probable cause, puts a legally enforceable burden on the entity possessing the records to produce the evidence regardless of where stored. The focus of the inquiry is on the legal location of the entity possessing the evidence, not the location of the evidence itself.

What this "solution" does not solve is situations where the jurisdiction where the evidence is physically located imposes limitations on access to the evidence that is inconsistent with the terms of the warrant from the investigating state. These situations arise in situations where data is stored in a foreign country, where presumably the MLAT

¹⁰⁵ *Searching and Seizing Computers, supra.*

¹⁰⁶ The Manual does, at pp. 84-86, discuss the jurisdictional issue (discussed above in this paper) but only in the context of determining in which judicial district a warrant should be sought.

¹⁰⁷ U.S. CONST. amend. IV.

¹⁰⁸ *United States v. Turner*, 770 F.2d 1508 (9th Cir.1985).

¹⁰⁹ *United States v. Nafziger*, 965 F.2d 213 (7th Cir.1992).

¹¹⁰ *Maryland v. Garrison*, 480 U.S. 79, 84, 107 S.Ct. 1013, 94 L.Ed.2d 72 (1987).

between the countries resolves the dispute. It may also arise, however, in situations where the required data is located in another state.¹¹¹

2.6.4 Conclusions as to Requirements for a Valid Warrant for Cloud Based Data

Obtaining a valid search warrant for cloud based data involves overcoming two obstacles largely unique to such data. The first is that the application for the warrant demonstrates probable cause to believe that the data sought is actually at the cloud location for which search is requested. The second is that, given the opaque and often proprietary processes that cloud providers use to allocate a user's data to the cloud servers, it may be difficult to meet the requirement that the warrant be reasonably precise in describing the location to be searched. Many of the issues in this obstacle are correlated to meeting the authenticity requirement for admission of cloud data, discussed in section 3.1.2, below.

2.7. Exceptions to the Warrant Requirement

There are exceptions to the rule that evidence obtained without a warrant is inadmissible. These are, of course, the "few, specific, established, and well-delineated exceptions" developed under the Fourth Amendment.¹¹²

Only two of these exceptions—consent and exigent circumstances—may justify the search of a cloud server without a warrant. These are discussed below.

It is important to remember that these exceptions simply allow the evidence to be used if it can be obtained. These two exceptions do not themselves provide authority for mandating a cloud provider to provide access.

2.7.1 Consent¹¹³

If the suspect or other person with authority over the property consents to a search, then a warrant is not required. In the context of cloud based data storage, the key questions for constitutionality analysis are: (1) who has the authority to consent, (2) is the consent given voluntarily and intelligently, and (3) is the consent limited?

As just noted, valid consent makes the acquisition of the data valid constitutionally. The cloud provider is not obligated to honor any consent other than that of the account holder. Nor is the cloud provider itself obligated to consent to a search, even if the provider is constitutionally capable of giving consent.

2.7.1.1 Who May Consent?

Obviously the suspect may consent to a search of his own premises and effects, subject to such consent being given voluntarily and intelligently. As for third party consent, the general rule is

¹¹¹ One of the authors, a former prosecutor, had an experience where a Washington warrant was served on a California Internet Service Provider. The California ISP insisted upon notifying the person whose records were being sought, to give them an opportunity to object, citing California law. The issue was ultimately resolved by obtaining a California warrant under the process referred to earlier as "domesticating the warrant," but the process took several months.

¹¹² *Schneekloth v. Bustamonte*, 412 U.S. 218, 219, 93 S.Ct. 2041, 36 L.Ed.2d 854 (1973), citing *Katz v. U.S.*, 389 U.S. at 357.

¹¹³ As noted earlier, the determination of whether there is a reasonable expectation of privacy in documents shared with third parties involves many of the same issues involved in a consent exception discussion.

that one “who possesses common authority over or other sufficient relationship to the premises or effects sought to be inspected” has the authority to consent to a search.¹¹⁴

The crucial analytical test is whether the defendant has a reasonable expectation of privacy in the premises or effects to which another has consented to be searched. In most instances of shared premises and effects the courts have ruled there is a reduced expectation of privacy in the premises or things shared with another.¹¹⁵ As the Ninth Circuit stated: “Although there is always the fond hope that a co-occupant will follow one’s known wishes, the risks remain. A defendant cannot expect sole exclusionary authority unless he lives alone, or at least has a special and private space within the joint residence.”¹¹⁶ The consent given by a third party may not be valid if the area involved has been set aside, or rented, for the exclusive use of the defendant.¹¹⁷

While consent issues in cloud stored data are not unique, consent issues will almost always arise because of the sharing of data with the cloud provider inherent in the cloud relationship. There are a wide variety of consent related issues involved in computer searches including when an employer can consent, when a parent or spouse can consent, etc. There is a rich body of law in these arenas that are applicable to cloud and non-cloud searches alike. Since our focus is on legally issues unique to or pervasive in cloud searches, the below discussion is limited to special instances of consent relevant to cloud based data, including the ultimate question of whether the cloud provider can consent to a search of a customer's data.

2.7.1.1.1. Parent

Courts generally find third-party consents easier to sustain “if the relationship between the parties—parent to child here, spouse to spouse in other cases—is especially close.”¹¹⁸

In the computer context, the more the computer is a shared item, both in terms of use and location, the more likely it is that a parent can give a valid consent. Even when the computer belongs to the child or is used by them exclusively and is located in their room or private space, parents can generally consent to a search of such a computer so long as the child is “essentially dependent” on the parent. If the child pays rent or in other ways is independent of the parent, the parent may lack the authority to give valid consent to a search of the child’s effects or private space.¹¹⁹

Parental consent is somewhat different in the cloud situation, because the physical storage location is not within the house. The crucial issue is likely going to involve shared access or ownership of the cloud *account*, not the local computer accessing the cloud-stored data..

2.7.1.1.2 Spouse

Spouses are generally considered to have joint control and equal right to occupancy of the premises and access to computers on the premises, and so may give a valid consent. If, however, the computer is used exclusively by the non-consenting spouse and is kept in a separate room (especially if locked), the other spouse may not be able to give valid consent. (See below section for discussion of encrypted or password protected files.)

¹¹⁴ *U.S. v. Matlock*, 415 U.S. 164, 171 (1974).

¹¹⁵ *E.g., U.S. v. Ladell*, 127 F.3d 622, 624 (7th Cir. 1997).

¹¹⁶ *U.S. v. Morning*, 64 F.3d 531, 536 (9th Cir. 1995), *cert. denied*, 516 U.S. 1152 (1996).

¹¹⁷ *State v. Mathe*, 688 P.2d 859 (Wash. 1984). Note that in Washington State, if a cohabitant with equal or greater authority to consent is present (whether or not the police know of their presence), that person’s consent must be obtained, notwithstanding the consent of a cohabitant with equal or lesser authority. *State v. Morse*, 123 P.3d 832 (Wash. 2005)/

¹¹⁸ *U.S. v. Ladell*, 127 F.3d 622, 624 (7th Cir. 1997).

¹¹⁹ *U.S. v. Rith*, 164 F.3d 1323 (10th Cir.), *cert. denied*, 528 U.S. 827 (1999).

The same analysis just made in the parental consent context applies here. The determinative issue is not shared occupancy and access to a local computer. It is instead shared ownership or access to the cloud account.

2.7.1.1.3. Co-user

Generally, allowing co-use results in a greatly reduced expectation of privacy. Whether password restricted or encrypted areas are the equivalent of locked bedrooms and thus not subject to a consent search by a co-user is a fact specific issue.¹²⁰ Under a 2007 4th Circuit decision, the use of password protected or encrypted files by the suspect prevents a co-user (including a spouse) from having actual authority to consent, but if the officer has no indication of the presence of password protected or encrypted files, the co-user may have apparent authority to consent.¹²¹ Until you learn otherwise, however, assume that a co-user can *not* give consent to search such objectively private areas of a computer.

For cloud computing the question is again, what authority over, and access to, the cloud account, as opposed to the computer, does the third party have?

2.7.1.1.4. Co-Tenants

Although the cloud providers system for storing user data may not physically segregate data by user, relying instead on logical segregation to track each user's data location, such intermingling should not create a consent issue regarding co-tenants as, unless specifically permitted by a user, users do not have access to each other's data.

2.7.1.1.5. Third Parties

When an item has been left in the custody of a third party the courts may find limits on the third party's authority to consent to a search of the item. In *United States v. James*,¹²² the Eighth Circuit held the third party did not have authority to consent to the search of a packet of computer CDs left in his possession by the defendant. The court focused on its perception of the defendant's intent and held that the defendant did not intend to give the third party any authority to consent to the search of the disks. He gave them solely for the purpose of storing the disks.

This view of a suspect's ability to limit the scope of authority given to a third party holding the defendant's item conflicts with decisions like *United States v. Falcon*.¹²³ The defendant left a cassette at his brother's apartment labeled "confidential" and "do not play". Notwithstanding these instructions, the *Falcon* court held the brother's consent to search the cassettes to be effective because the brother "could have played [the tape] at any time, whether by design or by accident."¹²⁴

One way of reconciling these two lines of thought is that in the James case the CDs were contained in a sealed envelope and the envelope was further encased in tape. Thus they could not have been viewed by accident. More importantly, the police came to third party in the James case

¹²⁰ In a very fact specific ruling the court in *U.S. v. Smith*, 27 F. Supp. 2d 1111 (C.D. Ill. 1998) allowed evidence obtained from a computer under a third party consent, noting that there was no evidence the defendant had password-protected the material seized.

The Fourth Circuit held that a search of an individual's password-protected computer files is an illegal search in violation of the Fourth Amendment when consent to the search was given by another individual who had joint access to the computer but did not know the passwords and therefore did not have authority to grant access to the files. *Truloch v. Freeh*, 275 F.3d 391, 403 (4th Cir. 2001).

¹²¹ *U.S. v. Buckner*, 473 F.3d 551, 555 (4th Cir. 2007)

¹²² 353 F.3d 606 (8th Cir. 2003).

¹²³ 766 F.2d 1469 (10th Cir. 1985).

¹²⁴ *Id.* at 1476.

because they intercepted a letter from James to the third party instructing the third party to destroy the envelope and its contents.

Whether the specific facts of *James* will form the basis of future limits on consent searches is unknown, but the case does provide support for the more widely held view that where the defendant has taken extra steps to insure the privacy of computer files, such as by encrypting or password protecting them, the authority of the third party possessing, or having access, to the computer containing those files to consent to the search of those files is greatly limited.

2.7.1.1.6. The Cloud Provider

The cloud provider is not acting as a parent or spouse. To the extent they have authority to consent to a search it would be because they are deemed a co-user, co-tenant, or third party with possession, with access authority sufficient to meet the law's requirements. It seems clear that the only thing that provides the necessary access authority is the terms of use contract, or some other authority given to the provider by the user outside the contract.

Clearly if the terms of use allow the provider to fully access or consent to another's access, that provides sufficient authority. The more important question is whether the terms of use authorize more limited access that a reviewing court would consider sufficient to allow the provider to consent to a search. The previous discussion at 2.3.1.1.2.6, regarding whether terms of use allowing the provider to mine the user's data for advertising purposes defeat a reasonable expectation of privacy is merely a different view of the same topic. In both situations the unique nature of the individual contracts and the lack of any case law in this arena make it impossible to predict the circumstances under which qualified access under the terms of agreement will be held to authorize the provider to consent to access by others including law enforcement. What can be said clearly is that the terms of use are highly relevant in this regard. It is doubtful that most users understand the exposure to third party access those terms can provide.

2.7.1.2. Consent May Be Limited and May Be Withdrawn

Consent may be limited and withdrawn. A search of areas for which consent was not given, or after consent is withdrawn, is invalid.¹²⁵

Some courts have held searches invalid when the officers, acting under general consent, broke open locked or sealed containers.¹²⁶ Is a search of hidden, password protected or encrypted files under a general consent to search a computer open to the same attack? To the extent that analogy to the physical world is applicable, the answer is probably no. The line of cases represented by *Wells*, focus on the destruction of property necessary to open the sealed or locked container. Since there is no comparable destruction in accessing hidden, password protected or encrypted files, the search may be upheld.

An interesting question, equally but perhaps not uniquely involved in cloud forensics, is what happens when someone with consent authority consents to a search of cloud data, the data is downloaded to a local police drive, and then, before the drive is actually examined, consent is withdrawn. May the police nonetheless search their drive containing an image or copy of the cloud data?

The authors have found no cases directly on point. In analogous situations, courts have held that where the IRS obtained tax documents by consent and then consent was subsequently withdrawn,

¹²⁵ See Model Code of Pre-Arrest Procedure § SS 240.3 (1975) (a consent search “shall not exceed, in duration or physical scope, the limits of the consent given”); *Mason v. Pulliam*, 557 F.2d 426 (5th Cir. 1977).

¹²⁶ *Florida v. Wells*, 539 So.2d 464 (Fla. 1989), *aff’d*, 495 U.S. 1 (1990).

the IRS must return, without further examination, any documents they have not examined, but may retain any incriminating documents they had already examined before the consent was withdrawn.¹²⁷

2.7.2 Exigent Circumstances

The exigent circumstances exception allows a warrantless search of premises where the officer has probable cause and exigent circumstances exist. The determination of whether exigent circumstances exist includes an examination of many factors.¹²⁸

We have long recognized that the imminent destruction of evidence may constitute an exigency excusing the failure to procure a warrant. This risk is particularly weighty where narcotics are involved, for it is commonly known that narcotics can be easily and quickly destroyed. ... We held that a police officer can show an objectively reasonable belief that contraband is being, or will be, destroyed within a home if he can show 1) a reasonable belief that third persons are inside a private dwelling, and 2) a reasonable belief that these third persons are aware of an investigatory stop or arrest of a confederate outside the premises, so that they might see a need to destroy evidence.¹²⁹

In the cloud context the most likely scenarios of exigent circumstances are where law enforcement is concerned that the defendant, or someone on her behalf, may delete cloud data before the data can be obtained under a warrant. This situation could arise where law enforcement does not have time to request/direct the cloud provider to “freeze” the data, or where the cloud provider is unwilling or unable to do so.

Assume the police are executing a lawful warrant and see, on an active computer screen, evidence that the computer is connected to a cloud server. Assuming they fear that an accomplice to the defendant may delete this cloud data before they can preserve it, may they download the cloud data to a portable police hard drive under a claim of exigent circumstances and then seek a warrant to search it?

A fascinating case that is the subject of a book by the AUSA prosecuting the case¹³⁰ was described by the court as follows:

¹²⁷ *United States v. Ward*, 576 F.2d 243, 244-45 (9th Cir.1978) and *Mason v. Pulliam*, 557 F.2d 426, 429 (5th Cir.1977).

¹²⁸ See *U.S. v. Brown*, 52 F.3d 415, 421 (2d Cir. 1995), *cert. denied*, 516 U.S. 1068 (1996) (“Our court has looked to six touchstones for determining the existence of exigent circumstances. Those are: ‘(1) the gravity or violent nature of the offense with which the suspect is to be charged; (2) whether the suspect is reasonably believed to be armed; (3) a clear showing of probable cause ... to believe that the suspect committed the crime; (4) strong reason to believe that the suspect is in the premises being entered; (5) a likelihood that the suspect will escape if not swiftly apprehended; and (6) the peaceful circumstances of the entry.’ Those factors are ‘merely illustrative, not exhaustive, and the presence or absence of any one factor is not conclusive.’” (citations omitted)).

U.S. v. Rico, 51 F.3d 495, 501 (5th Cir.), *cert. denied*, 516 U.S. 883 (1995) (“In evaluating whether exigent circumstances existed, we have found relevant the following factors: (1) the degree of urgency involved and amount of time necessary to obtain a warrant; (2) [the] reasonable belief that contraband is about to be removed; (3) the possibility of danger to the police officers guarding the site of contraband while a search warrant is sought; (4) information indicating the possessors of the contraband are aware that the police are on their trail; and (5) the ready destructibility of the contraband and the knowledge ‘that efforts to dispose of narcotics and to escape are characteristic behavior of persons engaged in the narcotics traffic.’” (citations omitted)).

¹²⁹ *U.S. v. Dawkins*, 17 F.3d 399, 405 (D.C. Cir. 1994).

¹³⁰ Schroeder, Steve (2011). *The Lure*, Cengage Learning.

Following an extensive national investigation of a series of computer hacker intrusions into the computer systems of businesses in the United States emanating from Russia, Alexey Ivanov was identified as one of the intruders. Around June, 2000, the FBI set up Invita, a “sting” computer security company in Seattle. On/about November 10, 2000 Mr. Ivanov, along with his “business partner,” Defendant Vasiliy Gorshkov, flew from Russia to SeaTac.

In Seattle, the two men met with undercover FBI agents at the Invita office located in Seattle. During the meeting and at the behest of the FBI, Defendant Gorshkov used an FBI IBM Thinkpad computer (“IBM”) ostensibly to demonstrate his computer hacking and computer security skills and to access his computer system, “tech.net.ru”, in Russia. After the meeting and demonstration, both Gorshkov and Ivanov were arrested.

Following the Defendants' arrest, without Defendant Gorshkov's knowledge or consent, the FBI searched and seized the IBM and all key strokes made by the Defendant while he used it, by means of a “sniffer” program which allowed the FBI to track and store the information. The FBI thereby obtained the Defendant's computer user name and password that he had used to access the Russian computer.

Armed with this information the FBI logged onto the subject computer(s) located in Russia. Faced with the possibility that a confederate of the defendant could destroy the files in the Russian computer, the FBI decided to download the file contents of the subject computer(s). This was done without reading same until after a search warrant was obtained.¹³¹

The court denied the defendant's motion to suppress this evidence under two theories. One, not relevant to this discussion, was the court's conclusion that the Fourth Amendment has no extraterritorial reach. The other justification was exigent circumstances—the agents had a legitimate fear that the data would be deleted before a warrant could be obtained. So long as the agents had probable cause for seizing (downloading) the data, and a valid warrant was obtained before the downloaded data was examined, there was no Fourth Amendment violation.¹³²

2.7.3 Conclusion as to Applicability of Exceptions to Warrant Requirement to Cloud-Stored Data

The consent exception is equally applicable to cloud data and may arise in most if not all instances because of the user's sharing of their data with the cloud providers and the possible or probable finding of consent implicit in such an arrangement. While this vitiates the warrant requirement, law enforcement is nonetheless dependent on voluntary cooperation from the cloud provider in gaining access to the data.

The exigent circumstances doctrine may be available for cloud stored data in situations where (1) law enforcement has probable cause to believe that evidence of a crime is stored in that data, and (2) they have some mechanism for downloading the data. The last requirement is true because without a warrant the cloud provider is not obligated to allow access even under exigent circumstances. Exigent circumstances excuses the absence of a warrant, but it does not provide for mandatory compliance.

¹³¹ *United States v. Gorshkov*, CR00-550C, 2001 WL 1024026 (W.D. Wash. May 23, 2001)

¹³² *Id.* at 3-4.

2.8 Conclusion Regarding Commanding Production

There are six issues relating to commanding production that are unique to, or provide special problems for, forensic examination of cloud stored data:

2.8.1 Does the sharing of data with a third party, inherent in cloud stored data, eliminate a reasonable expectation of privacy for the user (and the directly related consent issue stated in 2.8.5 below)?

We believe that it is unlikely that simply the act of storing data with a third party is not sufficient to defeat a claim of privacy. We also note that certain actions, such as encrypting the data can increase the likelihood that the courts will accept an argument that the user has a subjective expectation of privacy.

The two most important considerations are those expressed in the *Warshak* opinion discussed earlier, that email is akin to other forms of communication through third party intermediaries like phone conversations and mail and as such, is entitled to similar protection, but that the terms of the service agreement regarding access by the provider can be such that any privacy right is extinguished.

2.8.2 Does the SCA regulate disclosure by cloud providers?

We conclude that in some circumstance, where the cloud provider also provides electronic communication service, like Gmail, or where the cloud provider has no contractual right of access to the data, the SCA will apply. Outside those areas the SCA does not apply (but the reader is cautioned that the 4th Amendment may apply.)

2.8.3 Does cloud stored data create difficulties associated with meeting the requirement that there must be probable cause to believe the sought evidence will be at the location you seek to search and that a warrant be reasonably precise in describing the location to be searched?

These two questions revolve around the same stumbling block - the likely inability of the investigator to know precisely where (in a geographical or physical sense) in the cloud the sought for data is located. Without such information the affidavit can neither describe the location to be searched with any adequate degree of particularity nor state probable cause to believe the data will be at that location.

There are two possible solutions. The first is to use a subpoena mechanism to serve warrants, i.e., serving the document on the cloud provider's physical offices and requiring them to produce the requested documents wherever they are. This "solution" is limited to those states which allow a warrant to reach data located outside the state. It is further limited by complications associated with enforcing the warrant, and conflicting limitations on access to the data imposed by the state/nation where the data is located.

The second possible solution is not discussed in this paper for there is no data upon which to base an expectation of the solution. That solution is for the law to recognize, either in the form of changed rules and statutes, or in the form of case evolution, that brick and mortar concepts of location are not appropriate in these contexts and the real test should be on the adequacy of the logical separation of data between users and law enforcement's ability to articulate with specificity the logical location of data and the probability that evidence of a crime will be found at that logical location.

2.8.4 May the cloud provider consent to a search of cloud stored data?

As noted, this is in many ways, simply a different construct for the questions asked in the "reasonable expectation of privacy" inquiry and the answer is the same; the terms of the SLA and the degree of access, if any, given by the user to the service provider is largely determinative.

2.8.5 Are the consent and exigent circumstances exceptions to warrant requirements helpful if they permit but do not require disclosure by the cloud provider?

This question focuses not on large, constitutional issues, but on practical ones. If there are no constitutional prohibitions on disclosure, how does the investigator gain access to stored data? We conclude that SLAs, Privacy Policies, and the related expectations of the users will mean that providers will insist on being served with binding legal directives before they will disclose cloud data, even if there is no prohibition on the disclosure.

3.0 ACTUAL PRODUCTION IN COURT

Once obtained, evidence will be of little use if it is not actually accepted as evidence in a case—or accepted as evidence into a case. The Federal Rules of Evidence, or similar rules specific to the jurisdictional state, guide the process of admissibility. While there have been some modifications to these rules over the past few years to address the increasing role digital evidence is playing in the legal system¹³³, most decisions regarding the admissibility of digital evidence rely on rules originally designed for paper documents. Even more alarming is the fact that the legal education system has woefully failed to keep pace with these changes in the forms of evidence and lawyers and judges alike are often woefully unprepared to argue for, challenge, and rule on the admissibility of digital evidence.¹³⁴

The seminal court opinion in the 2007 case *Lorraine v. Markel American Insurance Company*¹³⁵ provides both a devastating critique at current practices in the digital evidence arena as well as a useful and comprehensive overview of the procedural requirements for admissibility of "Electronically Stored Information" (ESI) as guided by the Federal Rules of Evidence.

Because the general digital evidence legal arena is undeveloped, it is necessary to discuss the ways that digital evidence can be admitted or challenged before delving in to the unique problems presented by evidence obtained from cloud providers.

Judge Paul Grimm's opinion in *Lorraine*, with his exhaustive treatise of the rules governing admissibility of digital evidence, provides both a helpful road map to guide us through this area of the law and insight into what parts of the admissibility process are particularly problematic for evidence obtained from forensic examination of cloud based data.

¹³³ "E-Discovery Amendments to the Federal Rules of Civil Procedure Go Into Effect Today" *K&L Gates* (Dec 1, 2006). <http://www.ediscoverylaw.com/2006/12/articles/news-updates/ediscovery-amendments-to-the-federal-rules-of-civil-procedure-go-into-effect-today/>

¹³⁴ Alva, A. & Endicott-Popovsky, B. (2012). "Digital Evidence Education in Schools of Law." Paper presented at the ADFSLS Conference on Digital Forensics, Security and Law, Richmond, VA., Endicott-Popovsky, B. and Horowitz, D. (2012). "Unintended Consequences: Digital Evidence in Our Legal System", *IEEE Security & Privacy Magazine*. 10(2), 80-83.

¹³⁵ *Lorraine*, 241 F.R.D. 534.

Unlike the previous section which presented problems largely unique to the criminal investigatory process, this discussion of admissibility is largely applicable to both the civil and criminal process.¹³⁶

Before addressing the specific rules, it is important to note three things about the admissibility process. First, it is not stringently linear. As *Lorraine* states, there is a burden on opposing counsel to object to admissibility. Objections by counsel can be made based upon any one of the Rules of Evidence without regard to order.¹³⁷

Second, a judge's role in the admissibility process is to act as a gatekeeper. The bar for admissibility is generally low. Admitting evidence does not mean the evidence is persuasive or convincing. Admission simply acknowledges that the evidence can be presented to the jury and the jury will determine what weight to give the evidence. Judges will frequently acknowledge this by responding to an objection by saying "that goes to weight, not admissibility."¹³⁸ A common example of this in the ESI arena is an objection to the admissibility of a digital photograph on the grounds that it could have been altered. Absent any proof of alteration, such an objection goes to the weight the jury gives the evidence not its admissibility.

The importance of this two-step evaluation of evidence to a non-lawyer cannot be overemphasized. The first step is admissibility, the second step is persuasiveness. One way this could be viewed is to think of admissibility being the "weigh-in" of boxers before a match, where they must pass certain minimal but crucial eligibility and health requirements. Being allowed to enter the ring in no way guarantees success. But while the crucial, qualitative determination is made once the boxers are in the ring, where the goal is to win the boxing match, failure to meet the eligibility requirements to enter the ring is fatal.

The same is true in the courtroom. Getting evidence admitted does not guarantee a verdict in your favor, but failing to have the evidence qualify for admission can be fatal. In fact significant effort is frequently exerted in both civil and criminal cases in pretrial stages to qualify or disqualify evidence's admissibility. Often a victory or loss at the pretrial stage can set the ground for a settlement or guilty plea.

The third thing to note about the admissibility process is that trial judges have very wide discretion over admissibility and appellate courts generally defer to trial courts' determinations of admissibility. This puts an even greater emphasis on the determination of admissibility stage of the proceedings for trial court decisions regarding admissibility are only infrequently reversed on appellate review.

3.1 General Rules Applicable to all Electronic Evidence and Application to Cloud Context

Judge Grimm's discussion of the admissibility requirements of the evidence rules were in the context of email, but the rules discussed are generally applicable to all evidence, including other

¹³⁶ Two major differences between civil admissibility and criminal admissibility relate to the constitutional privilege against self incrimination and the constitutional right by the accused to confront accusatory witnesses. U.S. CONST. amends. V and VI. While these differences can be significant, the differences are generally not unique to cloud computing and are not discussed further other than noting their existence.

¹³⁷ *Lorraine*, 241 F.R.D. at 553

¹³⁸ Alas, there is a further complexity. This is explained in more detail by Judge Grimm in *Lorraine*, 241. F.R.D. at 539. , The simplest explanation is that because of the interplay between Fed. R. Evid 104(a) and (b), in addition to the judge, the jury must be persuaded as to the authenticity of an exhibit.

forms of digital evidence. He argues that five substantive areas for admissibility must be considered whenever Electronically Stored Information is offered as evidence. These are: relevance, authenticity, hearsay, original evidence, and probative v. prejudicial.¹³⁹ These categories are discussed in general below, and then in reference to unique problems associated with each category in cloud-based evidence.

3.1.1 Relevance

When ruling on admissibility, one of the first questions is whether the evidence presented has relevance: “(a) does it have the tendency to make a fact more or less probable than it would be without the evidence; and (b) is the fact of consequence in determining the action”¹⁴⁰. The bar for determining relevance is low, at “more probable than it would be without the evidence”¹⁴¹. Sufficiency is immaterial to admissibility based on relevance, as the evidence only needs to be shown to logically alter the probability of a fact of consequence.

Judge Grimm notes that establishing ESI as relevant is generally not difficult, although it is clear that “evidence that is not relevant is never admissible”¹⁴². Thus, relevance is generally not an issue.¹⁴³

3.1.1.2 Unique Cloud Issues – Relevance

The authors contend that there is nothing inherently unique to relevance with respect to the cloud.

3.1.2 Authenticity

To meet the authenticity requirement, the proponent must show that “the ESI is what it purports to be”.¹⁴⁴ The Federal Rules of Evidence provide non-comprehensive examples for how evidence can be authenticated:

- (1) Testimony of a Witness with Knowledge
- (2) Nonexpert Opinion About Handwriting
- (3) Comparison by an Expert Witness or the Trier of Fact
- (4) Distinctive Characteristics and the Like
- (5) Opinion About a Voice
- (6) Evidence About a Telephone Conversation.
- (7) Evidence About Public Records
- (8) Evidence About Ancient Documents or Data Compilations
- (9) Evidence About a Process or System
- (10) Methods Provided by a Statute or Rule.¹⁴⁵

¹³⁹ *Lorraine*, 241. F.R.D. at 538.

¹⁴⁰ Fed. R. Evid. 401

¹⁴¹ *Id.*

¹⁴² *Lorraine*, 241. F.R.D. at 541, citing Fed. R. Evid. 401

¹⁴³ Relevance can be an issue in resolving a hearsay objection. For example in a personal injury traffic accident case offer the issue of whether the defendant ran a red light might be relevant. If the plaintiff's attorney seeks to admit a letter from a third party containing that party's statement that the defendant ran a red light, the proper objection is “hearsay” because the third party is not present to be cross examined. If the proponent's response is that the letter is not being offered for the truth of its contents (and thus is not hearsay), the proper objection is not hearsay, but relevance. The letter itself, without regard to the truth of its contents is simply not relevant and thus inadmissible.

¹⁴⁴ *Lorraine*, 241. F.R.D. at 541-42.

¹⁴⁵ *Fed. R. Evid.* 901.

The two most relevant ways in which cloud-based evidence can be authenticated are through the testimony of a witness with knowledge (Fed. R. Evid. 901(b)(1)) and through evidence about a process or system (Fed. R. Evid. 901(b)(9)); both ways are discussed below.

3.1.2.1 Testimony of a Witness with Knowledge

In order to establish that evidence is authentic, the proponent will likely call a witness who can testify that the evidence is what it purports to be. "A witness may be appropriately knowledgeable through having participated in or observed the event reflected by the exhibit."¹⁴⁶

In the case of digital evidence, it is generally sufficient to produce a person, such as a custodian or other qualified witness, with personal knowledge of the procedure that generated the records.¹⁴⁷ Depending on the complexity of the system, the person may need more than a superficial knowledge of the system, however. As noted by Judge Grimm the witness must have a working knowledge of the process of producing the information:

[I]t is not required that the authenticating witness have personal knowledge of the making of a particular exhibit if he or she has personal knowledge of how that type of exhibit is routinely made. It is necessary, however, that the authenticating witness provide factual specificity about the process by which the electronically stored information is created, acquired, maintained, and preserved without alteration or change, or the process by which it is produced if the result of a system or process that does so, as opposed to boilerplate, conclusory statements that simply parrot the elements of the business record exception to the hearsay rule or public record exception. [Internal citations omitted.]¹⁴⁸

3.1.2.2 Evidence About a Process or System

If the proponent seeks to authenticate evidence by offering evidence about a process or system, the process/system itself may be subject to an additional test - the *Daubert* test. This test or standard, required to be conducted for cases in all U.S. Federal Courts, is used to determine admissibility of scientific, technical, or specialized evidence.¹⁴⁹ The *Daubert* test, complements Federal Rules of Evidence 702, which states:

A witness who is qualified as an expert by knowledge, skill, experience, training, or education may testify in the form of an opinion or otherwise if: (a) the expert's scientific, technical, or other specialized knowledge will help the trier of fact to understand the evidence or to determine a fact in issue; (b) the testimony is based on sufficient facts or data; (c) the testimony is the product of reliable principles and methods; and (d) the expert has reliably applied the principles and methods to the facts of the case.

Under *Daubert* the judge acts as a "gatekeeper", limiting such evidence to that which is reliable and scientifically valid. This determination is made by asking four questions:

¹⁴⁶ WEINSTEIN at § 901.03[2]

¹⁴⁷ *States v. Kassimu*, 188 Fed. Appx. 264, 2006 WL 1880335 (5th Cir. 2006)

¹⁴⁸ *Lorraine*, 241 F.R.D. at 545-46.

¹⁴⁹ *Daubert v. Merrell Dow Pharmaceuticals*, 509 U.S. 579, 113 S. Ct. 2786, 125 L. Ed. 2d 469 (1993). *Daubert* applied to scientific evidence. A subsequent case, *Kumho Tire Co., Ltd. v. Carmichael*, 526 U.S. 137, 119 S.Ct. 1167, 143 L.Ed. 2d 238 (1999), expanded this to include to areas of technical or specialized knowledge, and made it clear the test was flexible.

- (i) whether the proffered knowledge can or has been tested
- (ii) whether the theory or technique has been subjected to peer review and publication
- (iii) the known or potential rate of error, and
- (iv) whether the theory or technique has gained general acceptance in the relevant scientific discipline.

These four questions are often summarized as testing, peer review, error rates and acceptability in the relevant scientific community. Later court decisions have made it clear, however, that this should be a flexible test, tied to the particular facts of the case. These factors are neither an exhaustive list nor a required checklist.¹⁵⁰

3.1.2.3 Unique Cloud Issues – Authenticity

The main barrier for cloud forensics is one of authenticity. In particular, the challenge is to show that the data, both technically within the cloud environment and throughout the investigative process, is what it purports to be.

If the only authenticity challenge to cloud based evidence is whether the item produced in court is the item produced by the cloud provider the matter is akin to a chain of custody discussion, wherein all you must show is that what was provided by the cloud provider is what is being produced. Any number of witnesses can provide probative testimony, from a representative of the cloud provider who can compare the item presented in court to their record of what was produced, to a witness from the evidence proponent's staff who can testify from personal knowledge that what is produced in court was what he or she was given by the cloud provider. In fact this form of authenticity challenge is so easily met that it would rarely be raised.

The more meaningful authenticity challenge is to whether the item produced in court is in fact from the cloud storage records associated with a particular user. When evidence is seized from an individual's computer there can be little doubt that it came from that person's computer. To be sure, in a shared use environment the issue of what user actually used, possessed, created (or what ever other connection with the data is relevant) the data can be challenged, usually as a matter of weight, but there can be no sustainable challenge regarding what computer or hard drive was associated with the data.

In the cloud context, however, the data of multiple users is kept logically separate but is, by virtue of the nature of hard drives, physically comingled. If the authenticity challenge is to the proof that the data produced is the data associated with a particular user, some evidence or testimony must be produced to provide assurance that the thing is what it purports to be - data associated with a specific user.

Where the authenticity of a cloud-based document is challenged (i.e., is this document associated with the user's account), the person testifying to this will likely be a representative of the cloud provider who can testify about the process of associating data with specific users.

At one level this authenticity can be established by a witness from the cloud provider who can testify to the challenged fact, that the data offered was associated with a specific user when they produced it. This would be akin to a bank records custodian authenticating that a particular check was associated with a specific individual's account. A cloud provider witness would testify regarding their method of retrieving and producing the requested information ("I queried the

¹⁵⁰ *Kumho Tire*, 526 U.S. at 150, n. 25.

system for all records associated with a this user's account, copied that information onto a CD (or more realistically a hard drive) and delivered that CD/hard drive to the requestor.")

The more difficult question, and the more difficult authenticity proof requirement, arises when what is challenged is the core method of allocating, tracking and retrieving data associated with an individual account in the cloud. This is particularly true where mere possession of the item, like child pornography, with no other connection to the user is sufficient for conviction.

Meeting such a challenge requires proof of the reliability of the process or system controlling the data storage. The focus is on the reliability of the system rather than the reliability of the person describing the system.

Because of the newness, complexity, and likely proprietary nature of the cloud provider's file allocation procedures, it is highly likely an expert witness will be called upon to testify. The authenticity of the evidence gathered is at least partially dependent upon the qualifications and experience of the investigator-- a cloud provider employee in the case of cloud forensic investigations.

An even more onerous potential authentication problem arises related to how forensics is done in the cloud environment. Specifically, the rapid provisioning of resources within a single cloud environment creates a challenge to provide sufficient proof that the file(s) in question were actually those used, modified, or otherwise by the person in question. Cloud forensic investigations involve production of forensic evidence from the cloud provider, delivered to law enforcement. With traditional digital forensics, a potential authenticity issue within the chain of custody could be overcome because the evidence in question – the contents of the original hard drive can be physically presented to the court if necessary. Reexamination could be conducted with the physical evidence to provide additional proof of authenticity. With cloud forensic data, there is no fallback to physical evidence. Evidence presented for forensic investigations is reliant upon a proof that the process used by the cloud provider to attribute the questioned data to a particular user is reliable, for the original cloud servers would rarely, if ever, be available for in court examination.

The inability to ever produce the data in its original, as-stored-in-the-cloud-condition, makes the role of the process or system and the testimony about the process/system uniquely critical. Whether cloud providers will be willing to reveal this potentially highly proprietary information is an unknown.

3.1.3 Hearsay

Hearsay has been described as

an out-of-court statement offered in court to prove the truth of the matter asserted by the out-of-court declarant. It is offered into evidence through the testimony of a witness to that statement or through a written account by the declarant. The hearsay rule excludes such evidence because it possesses the testimonial dangers of perception, memory, sincerity, and ambiguity that cannot be tested through oath and cross-examination.¹⁵¹

The rule favors live testimony where such would help the fact finder assess the veracity and perception of the person making a statement being offered for its truth.

¹⁵¹ Rice, P. (2005) *ELECTRONIC EVIDENCE: LAW AND PRACTICE*. ABA Publishing.

In general, where a statement is offered for its truth, the person making the statement must testify. A classic example of a hearsay statement is a witness in an automobile accident case testifying that the police officer investigating the accident told him (the witness) the defendant was speeding. If the officer is not present, then she cannot be cross-examined about the basis for her statement. Since the statement of the officer, given by the witness, is likely only relevant for the truth of what the statement asserts (the defendant was in fact speeding) the witness' testimony about what the officer said is clearly inadmissible hearsay.

Hearsay is an extremely complex area of evidence law and is the subject itself of many learned treatises and books. We can not even begin to explore the intricacies of this subject in this chapter. A reader, uninitiated in this subject, would be well served by referring to the often cited Judge Grimm and his decision in *Lorraine v. Markel American* for an excellent summary of hearsay law in the context of ESI.¹⁵²

Judge Grimm summarized the hearsay rule as requiring the application of five separate questions:

- (1) Does the evidence constitute a statement, as defined by Rule 801(a); (2) was the statement made by a "declarant," as defined by Rule 801(b); (3) is the statement being offered to prove the truth of its contents, as provided by Rule 801(c); (4) is the statement excluded from the definition of hearsay by rule 801(1); and (5) if the statement is hearsay, is it covered by one of the exceptions identified at Rules 803, 804 or 807.¹⁵³

Some of Judge Grimm's assessments of the application of the hearsay rule to ESI are worth repeating.

Electronic evidence generated by a computer process (log files for example) is not hearsay because they are not 'statements' "¹⁵⁴, with a statement being "a person's oral assertion, written assertion, or nonverbal conduct, if the person intended it to be an assertion" "¹⁵⁵.

Such electronic evidence would also not be considered a statement made by a declarant (under the given definitions for "person" and "declarant" from Fed. R. Evid 801) because "when an electronically generated record is entirely the product of the functioning of a computerized system or process... there is no 'person' involved in the creation of the record, and no 'assertion' being made. For that reason, the record is not a statement and cannot be hearsay" "¹⁵⁶. It is important to note that this is only for electronic records produced by a system or process, and does not directly apply to electronic writings where a person makes an assertion in his or her statement.

Several important hearsay exceptions—including business and public records—reinforce the importance of authentication. The business records exception and others can allow admission provided that it meets the proper qualification and that "neither the source of information nor the method or circumstances of preparation indicate a lack of trustworthiness." "¹⁵⁷

¹⁵² *Lorraine*, 241 F.R.D. at 562, *et. seq.*

¹⁵³ *Id* at 562-63.

¹⁵⁴ *Lorraine*, 241 F.R.D. at 564; *U.S. v. Khorozian* 333 F.3d 498, 61 Fed. R. Evid. Serv. 980, 2003.

¹⁵⁵ Fed. R. Evid. 801(a)

¹⁵⁶ *Lorraine*, 241 F.R.D. at 564-65.

¹⁵⁷ Fed. R. Evid 803(6)(E)

3.1.3.1 Unique Cloud Issues – Hearsay

Most of the hearsay analysis applicable to ESI in general will be applicable to ESI stored in the cloud. The five-step analysis prescribed by Judge Grimm must be followed whether the data being offered comes from the defendant's local hard drive or from cloud-based storage. The one area where hearsay rules are more frequently applicable in the cloud context is in establishing the authenticity of the cloud-based data (with authenticity meaning producing evidence that the data in question is data associated with the relevant user account.)

We have previously described, at 3.1.2.2, how such authenticity could be established by showing the process or system involved in attributing data to a particular account. A proponent of an exhibit obtained from cloud storage might seek to authenticate an exhibit without calling a witness from the cloud provider, arguing that a written certification that the exhibit was a record of regularly conducted activity was sufficient under Fed. R. Evid 902(11). That is, the cloud provider would provide a certification stating that the exhibit was stored in a location attributable to the user in question at or near the time it was transmitted to the cloud provider by the user, that it was maintained in that location in the course of the regularly conducted activity of the provider, and that it is the regular practice of the provider to conduct this activity.

It seems to stretch the business records exception beyond any logical meaning to allow it to be used in this manner, to avoid providing the proprietary details of the storage and allocation system, but it is not a stretch to believe that some enterprising attorney, faced with a recalcitrant cloud provider unwilling to provide proprietary details, might attempt such use.

A more likely use of the business records exception would be to use that exception, together with the argument that log files of customer access are statements of a computer, not a person, as a way of admitting such log files without calling a witness to authenticate the log files.

This again emphasizes the importance of authentication and methods of meeting authentication objections in the cloud environment.

3.1.4 Original Evidence

The Original Writing Rule is the next hurdle Judge Grimm sets out for electronic evidence admissibility. Fed R. Evid. 1002 requires the proponent offer the “original” of a writing, recording, or photograph unless the Rules provide otherwise. Fed. R. Evid. 1003 states that “A duplicate is admissible to the same extent as the original unless a genuine question is raised about the original’s authenticity or the circumstances make it unfair to admit the duplicate.”

Offering a duplicate or copy of an original, thus, is generally only a problem when there is a genuine question original’s authenticity.

3.1.4.1 Unique Cloud Issues – Original Evidence

Any unique cloud issues relating to the original writing rule turn on the authenticity questions discussed earlier. So long as the proponent can show that the offered item is what it purports to be (which in the cloud context means that the document offered is a copy of information that can be shown to have come from an area of the cloud assigned to the defendant) there should be few original evidence issues with regard to cloud based data.

As with all the other admissibility questions there could easily be obstacles to the admission of documents stored in the cloud related to the documents themselves and not to the fact they were stored in the cloud.

The challenges of admissibility with respect to the unique nature of the cloud stem primarily from the fact that the “original” evidence – the multitude of hard drives containing the data attributable to the defendant – will almost never be available. The inability to ‘go back’ and obtain the original again is a unique issue that presents challenges for cloud forensic investigations from an authenticity standpoint.

The unique requirement placed upon actual production is that cloud providers conduct the investigation. This requires that cloud providers comply with actual production requests. Would inability to access original evidence halt an investigation? Federal Rules of Evidence 1004 states that an original is not required if, among other conditions, (b) an original cannot be obtained by any available judicial process; or (c) the party against whom the original would be offered had control of the original; was at that time put on notice, by pleadings or otherwise, that the original would be a subject of proof at the trial or hearing; and fails to produce it at the trial or hearing”. Thus there may be a potential to gather evidence on the surface, though no accompanying metadata, logs, or other data indicating proper authenticity would be available. Depending on the case and evidence sought, production may not be possible. If cloud providers did not comply with production, the authors note that the burden in determining authenticity of any evidence collected through other means would be higher, and potentially unattainable.

3.1.5 Probative versus Prejudicial

The final admissibility question noted by Judge Grimm is probative value against the potential for unfair prejudice. This admissibility test is governed by the Federal Rules of Evidence 403, which protects against unfair prejudice. When counsel “analyzes the admissibility of electronic evidence, he or she should consider whether it would unfairly prejudice the party against whom it is offered, confuse or mislead the jury, unduly delay the trial of the case, or interject collateral matters into the case.”¹⁵⁸ With ESI evidence, Judge Grimm states that the courts are likely to consider whether the evidence would be unduly prejudicial for several reasons including when there is a concern as to the “reliability or accuracy of the information that is contained within the electronic evidence”.¹⁵⁹

3.1.5.1 Unique Cloud Issues – Probative versus Prejudicial

This admissibility question will apply equally to cloud and non cloud-based evidence. It might be possible for a party to seek to restrict reference to the fact that data was stored in the cloud if that fact might be prejudicial, but the authors are unaware of a case where this argument has been made.

3.2 Conclusions Regarding Admissibility of Cloud Based Data

3.2.1 Authenticity

As emphasized, authenticity is likely the most critical admission criteria unique to the cloud. Because individual user data is almost always comingled physically it will be necessary to establish the validity of the process of attributing specific data to specific users. The unique aspects of cloud-based evidence present challenges in authenticity that must be considered

¹⁵⁸ *Lorraine*, 241 F.R.D. at 584.

¹⁵⁹ *Id*; *St. Clair v. Johnny's Oyster & Shrimp, Inc.*, 76 F. Supp. 2d 773, 774 (S.D. Tex. 1999)

throughout a cloud forensic investigation. Judge Grimm's opinion strongly advocated diligence in evidence authentication: "the inability to get evidence admitted because of a failure to authenticate it almost always is a self-inflicted injury which can be avoided by thoughtful advance preparation"¹⁶⁰. A technical effort to certify systems are forensically ready; a well-documented investigative process; and a clear legal argument for admissibility would be indicative of 'thoughtful advanced preparation' required to overcome the unique barriers for the authentication and admissibility of cloud forensic evidence.

Even with such a system in place, those seeking to admit cloud data are dependent upon the provider to produce the witness(es) necessary to describe the system.

The ultimate answer to questions involving authenticity of cloud data will turn, thus, on the validity of the system used to logically allocate data to individual users, and the ability of the proponent of such evidence to find a persuasive way to describe such a system. This will involve the following questions as a summary of authenticity issues for potential cloud based evidence:

3.2.1.1 Was the Process Used to Attribute Data to User Reliable?

The actual production of evidence will be carried out through a warrant or subpoena provided to a cloud provider. An employee of the cloud provider would then produce the evidence and present findings. While forensic investigations are allowed to occur without the presence of law enforcement,¹⁶¹ the validity of the investigation and the processes used to gather the evidence are subject to proper authentication. How do we know the evidence gathered by a cloud provider is what it purports to be? What proof is there than the process used by the cloud provider to attribute evidence to a particular user is reliable?

The two most meaningful ways to show authenticity under the Federal Rules of Evidence are 1) Evidence About a Process or System, and 2) Testimony of a Witness with Knowledge.¹⁶²

In both options, a representative from the cloud provider would likely be necessary to connect the evidence in question to the cloud customer.

3.2.1.2 Was the Process Used to Obtain Forensic Data from the Cloud Reliable?

At its most basic level proof of authenticity for cloud based data is proving that the data is what it purports to be – data from the cloud storage area associated with a specific person. The most convincing way to do this is to show that process used to extract cloud data (and by necessity, the process used to allocate and track the location of data associated with a specific person) is reliable. This can be done by describing the results of testing the process, by describing the logic behind the process, and by having expert testimony about the reliability of the process. The newness, complexity, and likely propriety nature of the cloud provider's data allocation procedures require the cooperation of the cloud provider and establishing reliability may, at least until the process is widely accepted as reliable, be a formidable task.

3.2.1.3 Can the Cloud Forensics Process Pass a *Daubert* Test?

¹⁶⁰ *Lorraine*, 241 F.R.D at 542.

¹⁶¹ *Bach*, 310 F.2d 1063,

¹⁶² Fed. R. Evid. 901

The *Daubert*¹⁶³ test sets a higher bar for authenticity, and is an additional measure to validate the process used to obtain forensic data and other authenticity questions. This test is required in U.S. Federal Cases for scientific, technical or other evidence requiring specialized knowledge, including digital evidence. There are four lines of inquiry that an expert witness is questioned on for the evidence in question: i) whether the proffered knowledge can or has been tested; ii) whether the theory or technique has been subjected to peer review and publication; iii) the known or potential rate of error; and iv) whether the theory or technique has gained general acceptance in the relevant scientific discipline.

The requirements described by the *Daubert* test provide a charge to the academic and industry community of cloud forensics researchers and practitioners. Beyond all other legal requirements that must be considered, one crucial gate by which cloud forensic evidence would be admitted in a court case rests on reviewed and published techniques; knowledge of the rate of shortcomings; and acceptance by the cloud forensics community of the processes used to attribute data in the cloud to an individual user. For an emerging area, this monumental, but necessary, challenge calls for a cohesive 'best practices' approach motivating the field of cloud forensics such that evidence from cloud forensic investigations can be used in court.

3.2.1.4 Is the use of Duplicates Sufficient? Is the Inability to Access the Underlying Original Data Fatal to establishing Authenticity?

Where a local drive is the subject of forensic analysis, one unspoken fact usually undercuts and deters most challenges to the accuracy of the forensics process - regardless of how the data was found, the original drive is usually available to demonstrate that the data is in fact on that drive. While examination of the original drive in this manner would be a rare act, the potential to do so is always present. The existence of the possibility can deter many theoretical challenges.

The absence of such underlying original data will likely not be fatal to a legal challenge where the issue is being decided by a judge, but where the jury is the ultimate determiner of relevance, which is in itself dependent on a piece of evidence being what it purports to be, the inability to produce the original evidence may be faced with lay skepticism. The jury is being asked on faith to accept the validity of the process used to obtain the evidence.

The problem with the absence of the original underlying data is difficult enough where the evidence copy has been obtained through the forensic practices of the provider. When the copy is acquired without the use of acceptable forensic practices, in this case due to access issues, the difficulty of establishing authenticity may be insurmountable.

4. ADDITIONAL LEGAL REQUIREMENTS

This section will detail commonly cited legal issues for cloud forensics, and will discuss whether or not each issue presents a unique legal challenge with respect to the cloud. These legal requirements should be considered throughout cloud forensic investigations.

4.1 Multi-Jurisdiction

Multi-jurisdictional issues are consistently noted as the primary issue in cloud forensic investigations, and digital forensic investigations as a whole.¹⁶⁴ The location of data affects the

¹⁶³ *Daubert*, 509 U.S. 579.

¹⁶⁴ Ruan, K., Carthy, J., & Kechadi, T. (2011a). *Survey on cloud forensics and critical criteria for cloud forensic capability: A preliminary analysis*. Paper presented at the 6th annual conference of the

ability to compel production of such data and may, although unlikely under most states' long arm jurisdiction rules, affect the determination of where a case involving cloud data must be filed/prosecuted.

4.1.1 Criminal Cases

Criminal cases involve criminal charges brought by the government under criminal sections of the U.S. Code or state counterparts. The jurisdictional issue for where such a case may be filed usually turns on questions of where the victim is located, where the defendant is located, and where the criminal acts occurred or where their impact was felt. Often jurisdiction is not exclusive, i.e., several states or the federal government and one or more states may each legitimately assert jurisdiction. Subject to double jeopardy limitations prosecution in multiple jurisdictions is possible.

In general, jurisdiction for a case will be in the jurisdiction investigating and filing the case, although this is largely a tautological decision for usually the selection of what jurisdiction will investigate a case is decided by the determination of which jurisdiction may prosecute the case. In cases with multiple possible jurisdictions, so long as the valid statutory jurisdictional requirements are met, the ultimate decision as to which will prosecute is usually a policy question.

For criminal cases, the controlling substantive, procedural and constitutional rules (including the Rules of Evidence) are those of the government asserting jurisdiction and bringing the case.

4.1.2 Civil Cases

Cases where one person or entity brings a claim against another person or entity for a failure of a legal duty are considered civil cases. The government can be a party, either as plaintiff or defendant; the key requirement is that the cause of action is civil. For civil cases, an understanding of jurisdiction is more complicated. When the data and entities involved in the case are in different geographic areas, the primary jurisdictional requirement is that the “forum” state (the state where an action is brought) has “enough connection with a problem to satisfy constitutional and statutory requirements”¹⁶⁵. Some courts contend that location of a server is not sufficient enough to qualify as a connection for jurisdiction decisions.

Issues other than determining jurisdiction can be involved in multi-jurisdictional cases. One significant issue is whose substantive law applies when the parties and evidence are located in different jurisdictions? A large and complex area of law called “Conflict of Law” has been developed to resolve these issues. This body of law is beyond the scope of this article and will not be addressed here.

The most important point to be made for this article is that the issues associated with the multi-jurisdictional nature of cloud based data storage are not unique. These issues arise frequently in disputes completely outside the realm of cloud storage.

4.2 Multi-Tenancy

ADFSL Conference on Digital Forensics, Security and Law, Richmond, Virginia, USA (p. 9).
Liles, S., Rogers, M., & Hoebich, M. (2009). A survey of the legal issues facing digital forensic experts.
Advances in Digital Forensics V, (pp. 267–276), Springer.

¹⁶⁵ Richman, W. M., & Reynolds, W. L. (2002). *Understanding conflict of laws* (p. 428). Lexis Nexis Matthew Bender.

Multi-Tenancy issues are endemic to cloud forensic investigations due to the shared storage nature of cloud computing. There are two issues, each already discussed in the previous two sections. The first is the validity-of-the-warrant issues relating to establishing probable cause to believe that evidence of a specified crime will be found at the location for which search permission is sought, and the related issue of the need for particularity in the warrant regarding the identification of the place to be searched.

The second issue is authenticity. If data from multiple tenants is stored at the location to be searched, there must be a sufficient basis for claiming that the offered data is that of the defendant and only the defendant.

Where the investigator cannot specify with precision the location of the sought after data there may be temptation to seek “cloud-wide” warrants or other compulsory production orders. Such over broad orders may look to a reviewing court like a fishing expedition. Such overbroad, general warrants are unlawful searches, and the results of such searches will almost certainly be suppressed.

Such a broad warrant, one that does not limit itself to a specific user's data, can be overbroad in two ways. First, such a warrant almost by definition is reaching or has the potential to reach the data from tenants with no involvement in the matter before the court. This could conceivably rise to a trespass or invasion of privacy action against the party seeking the evidence. Second, it could involve reaching data of the target outside the scope of the warrant or subpoena. Either of these flaws can lead to suppression.

4.3 Service Level Agreements (SLA)

Service level agreements (SLAs) govern the relationship between the customer and the cloud service provider. As such, the terms agreed to within the SLA may provide information on how forensic investigations will be handled. A large majority of cloud forensics survey participants noted that tools, techniques and other information for forensics investigations should be included in SLAs.¹⁶⁶ While a review of current SLAs is outside the scope of this chapter, it is important to provide a brief overview on the legal implications of SLAs and how SLAs effect investigations.

Legally, SLAs are almost always binding fall under U.S. contract law. SLAs can be of importance particularly when setting terms for collection of forensic data. The most common burden SLAs place on the cloud provider are with respect to uptime for the customer, though several have advocated that these agreements include how incidents will be handled,¹⁶⁷ including the processes for conducting investigations that respect the laws of multiple jurisdictions.¹⁶⁸

From an investigative standpoint, the SLA dictates the availability of forensic data for the customer that could be collected in the event of an investigation. While much of the focus of this chapter has related to obtaining cloud data in an adverse relationship (i.e., the government seeking the defendant's cloud data), in many instances the cloud data at issue belongs to the victim, and the issue is not how to command such production but rather the rights the user has to her own forensically accurate data. If the SLA does not include notice of what type of process or forensic data will be provided for the customer, then the cloud provider has no contractual duty to

¹⁶⁶ Ruan, K., Carthy, J., & Kechadi, T., 2011a (pp. 11-13)

¹⁶⁷ Grobauer, B., & Schreck, T. (2010). Towards incident handling in the cloud: challenges and approaches. *Proceedings of the 2010 ACM workshop on Cloud computing security workshop*, 77–86. doi:10.1145/1866835.1866850

¹⁶⁸ Ruan, Carthy, Kechadi, & Crosbie, 2011b (p. 43)

provide such information. This does a couple of things legally: 1) it binds access to forensic data that may otherwise be available; 2) lowers quality of best evidence available. The SLA may govern what type of forensic data is collected and the process in which it is stored.

It is important to note that an SLA is only binding between the parties and does not restrict the information that may be sought under a warrant or subpoena. An SLA that denies any provider responsibility to give a user log files is not a shield against a warrant asking for log files. If they exist they must be produced.

4.4 Chain of Custody

A crucial component of admissibility of evidence in court is whether a well-documented and validated chain of custody is maintained for such evidence.¹⁶⁹ When evidence is “susceptible to alteration by tampering or contamination,” then a “substantially more elaborate foundation” may be required.¹⁷⁰ There are many factors that present chain of custody concerns for cloud forensic investigation that would introduce susceptibility, and thus a more rigorous requirement for a proper chain of evidence.

The first potential failure of the chain is with the cloud provider. At this point, as we asserted earlier, forensic evidence will be obtained by the cloud provider and presented to law enforcement as evidence. With this reality, there is no control on the forensic investigation with respect to procedure, process, or person; the collection of evidence is conducted ‘behind doors.’ While it is allowed under the Fourth Amendment for search and seizure without the presence of law enforcement,¹⁷¹ diligence must still be conducted for a proper chain of evidence. The burden of documenting the chain of custody rests on the cloud provider, and such documentation is critical to ensuring the chain is maintained. Detailed documentation should include the person conducting the investigation, the steps taken to ensure the evidence has not been modified, and verification through hashes.

With regards to the person—particularly when there are multiple persons—conducting the investigation, a proper chain of evidence must be maintained through logs and comprehensive notes to detail who conducted what elements of the investigation as well as how the evidence was handed off and stored securely. Additionally, establishing the person(s) on the cloud provider side had familiarity with the investigation process used by the company may be a useful step in admissibility for authentication.¹⁷² This would be inline with authentication through expert witness testimony as previously discussed.

Ensuring the evidence has not been modified also must be established. A proper process and inclusion of verification through hashes should be conducted and documented to confirm evidence had not been modified, which will stand as verification in court.¹⁷³ However this may be

¹⁶⁹ Kuntze, N., Rudolph, C., Alva, A., et. al, 2012

¹⁷⁰ Losavio, M. (2005). The law of possession of digital objects: Dominion and control issues for digital forensics investigations and prosecutions. *First International Workshop on Systematic Approaches to Digital Forensic Engineering*, 2005. 177–183.

United States v. Miller, 994 F.2d 441 (8th Cir. 1993) (discussing chain of custody concerns)

¹⁷¹ *Bach*, 310 F.2d 1063, for example.

¹⁷² *United States v. Cameron* 762 F.Supp.2d 152, 158-159 (D. Maine 2011)

¹⁷³ Jarrett, H. M., Bailie, M. W., Hagen, E., & Judish, N. (2009). Searching and Seizing Computers and Obtaining Electronic Evidence in Criminal Investigations. *OLE Litigation Series*. Retrieved from <http://www.justice.gov/criminal/cybercrime/docs/ssmanual2009.pdf>

a technical problem in cloud forensic investigations where cloud images may not be able to be validated using cryptographic hashes.¹⁷⁴

Another issue with ensuring a proper chain of evidence is that many cloud providers use proprietary file systems for provided services. This introduces questions of validity, and presents a gap in familiar digital forensics practices handling hard drives.

5.0 FUTURE RESEARCH DIRECTIONS

To our knowledge, this is the first comprehensive work on the legal process and requirements for cloud forensics and cloud-based investigations. Our work in this chapter has established a foundation for further research in the legal aspects of cloud forensics. We plan to build on this work through several research directions.

From the legal perspective, there is an alarming gap in the understanding of digital evidence—including emerging areas such as cloud-based evidence. This lack of education for members of the legal community can result in serious miscarriages of justice and disruption of the legal system.¹⁷⁵ We actively plan to extend this research by identifying necessary elements that must be taught in schools of law, and to current members of the judiciary. This includes the development of curriculum to include interdisciplinary courses that will improve digital evidence literacy among law students and educate on evidentiary rules relevant in cloud forensics cases. As the cloud becomes more prevalent, we will begin to see case law develop around how cloud-based evidence is handled. We will expand our research to include a thorough analysis of new cases from the courts in this area. Such research would inform the legal and technical communities of potentially new legal requirements set by the courts, and also provide any new guidance for how to approach cloud forensics.

The area of cloud forensics must include collaboration between the technical, organizational, and legal perspectives.¹⁷⁶ This work sets a firm legal foundation that must be combined with other areas to build multidisciplinary solutions. We plan to conduct research to combine legal requirements with technical solutions to ensure cloud computing environments are forensically ready. There is a need for best practices and guides for cloud providers, cloud customers, and other stakeholders to understand the process and requirements we detail in this work, which we will seek to provide through collaborations. We also plan to research the legal barriers for eDiscovery of data stored in the cloud, including the economic costs that may be incurred by businesses for cloud forensic investigations.

Additionally, there are areas related to cloud forensics not detailed in this work. Such areas include specifics on how network data is handled in court cases; the differences between the acquisitions of physical disks and cloud data; and the legal considerations on the storage of forensic data in the cloud with regard to providence and a valid chain of custody. These are all potential avenues for further research that should be sought to extend the field.

And of course, there is the expansion of the analysis contained in this chapter to the civil arena, particularly in regard to the processes of obtaining cloud data.

¹⁷⁴ Dykstra, J., & Sherman, A. T. (2011). *Acquiring Forensic Evidence from Infrastructure-as-a-Service Cloud Computing: Exploring and Evaluating Tools, Trust, and Techniques*. Paper presented at the DFRWS Conference. New Orleans, LA.

¹⁷⁵ Alva & Endicott-Popovsky, 2012 . Endicott-Popovsky & Horowitz, 2012 (pp. 80-83)

¹⁷⁶ Ruan, Carthy, Kechadi, & Crosbie, 2011b (pp. 37-40)

Overall, there are many avenues for research that must be explored in order for the emerging field to develop reputable findings generally accepted by the field, in turn, be used to aid in the admissibility of cloud-based evidence in courts of law.

6.0 CONCLUSION

This work has provided insight on the legal process for cloud forensic investigations. This process includes the constitutional and statutory limitations determine level of protection provided, and the level of warrant or subpoena that may be necessary to conduct an investigation. A key focus was on the terms of SLAs regarding access to data granted by the user to the provider. The admissibility process, following actual production, includes the legal requirements necessary to ensure evidence obtained from the cloud passes admissibility, particularly authenticity. From the proceeding summary of insights, it is possible to develop a set of requirements that can be incorporated into system design that will render cloud computing environments forensically ready—prepared to collect admissible evidence that will survive technical challenges. In the cloud environment, with little ability of the user to control forensic readiness, aside from contractually, this takes the form of contractual obligations demanded by users of their cloud provider, audit certifications from cloud providers documenting the validity of their forensic readiness program and procedures and storage on the part of users who will need to retrieve this evidence on demand. The authors expect that this set of processes and requirements can be reduced to a series of technical requirements that will “bake-in” the qualities of forensic readiness and render more likely the ability of users to rely on cloud providers to collect and store valid and admissible evidence.

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